Research Master Thesis

Exploring Monotonic Reasoning in Neural Natural Language Inference

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a thesis submitted in partial fulfilment of the requirements for the degree of

MA Linguistics

(Human Language Technology)

Vrije Universiteit Amsterdam

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Submitted: June 17, 2026

Declaration of Authorship

I, author, declare that this thesis, titled *Exploring Monotonic Reasoning in Neural Natural Language Inference* and the work presented in it are my own. I confirm that:

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Date: 17.8.2026

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Acknowledgments

First and foremost, I would like to thank my supervisors, Dr. Antske Fokkens and Leon Eshuis for all their support and devotion throughout this project. I am really grateful not only for their valuable feedback and guidance, but also for all the interesting conversations, exchange of ideas, and perspectives we shared this past year. I feel really fortunate that I had the opportunity to work with and learn from both of them.

l would also like to thank Dr. Luis Morgado da Costa, for his mentorship throughout this Master's. I deeply appreciate that he always made time to help me and share his advice whenever I needed it. I would like to extend my thanks to all the professors and members of the CLTL Lab for introducing me to a field I knew very little about, but have come to find incredibly interesting.

Many thanks to my fellow students Ariana Britez, Farnaz Bani Fatemi, Szabolcs Pál, Csenge Szabó and Ruth Scheichl for all the time we spent working together and all the memorable moments we shared.

I am more than grateful to my parents, who have always tried with heart and soul to give me the best education possible, for always believing in me and for putting up with my overthinking for 27 consecutive years - something that I couldn't help but inflict on this thesis as well. I would also like to thank my sister and my best friends, who have always been there for me, even when we were apart. Finally, I want to thank Ioanna, my childhood friend, roommate, and best pastry chef in Amsterdam, for being an endless source of support, joy and freshly-baked chocolate chip cookies; but most importantly for teaching me to not cry over spilled milk - or in my case, over spilled code.

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Abstract

This thesis investigates the ability of two Transformer models, GPT-2 and BERT-base, in *Monotonicity Reasoning*, a highly systematic type of logical deduction and a fundamental component of Natural Language Inference (NLI). Despite the successes of neural NLI models on general benchmarks, recent work has shown that LLM inference is often driven by spurious strategies. One increasingly popular approach for interpreting these shortcomings is the development of phenomenon-specific datasets, aiming to show whether models trained on high-level objectives leverage appropriate task properties. To this end, we introduce a new diagnostic dataset, *Monotonic Reasoning* (MoRe) that targets the phenomenon of Monotonic Reasoning in Natural Language Inference (NLI). Our methodology is centered around a series of controlled experimental splits, where we carefully manipulate two key components of monotonic logic, namely the semantic relationship of *hypernymy* (e.g. *beagle* $\sqsubset$ *dog*) and its interaction with *logical operators* (e.g. *all, some, no*). These evaluations allow us to investigate the extent to which our models encode and rely on these particular linguistic features when generalizing across different train/test splits, as well as help us diagnose potential shortcut strategies underlying their predictions. Through our experiments, we provide useful evidence on how models handle linguistic and logical properties within monotonicity, and highlight broader strengths and weaknesses of LLM inference.

Chapter 1

Introduction

Large Language Models, following the introduction of the Transformer architecture Vaswani et al. (2023), have demonstrated impressive progress across a wide range of Natural Language Processing tasks and benchmarks (Brown et al., 2020; Liu et al., 2019; Wang et al., 2019a). Large-scale pre-training and the resulting versatility of the so-called *foundation models* (FMs) (Bommasani et al., 2021) have established them as the underlying technology behind various applications, including question-answering (Wan et al., 2025), information retrieval (Lewis et al., 2021), and conversational AI (Ouyang et al., 2022), raising both public and scientific expectations regarding what these systems are capable of achieving and how far scaling laws may ultimately take them.

Despite these advances, the nature of LLMs ' abilities remains a subject of ongoing debate. Part of the problem lies in the opaque design of these models. LLMs are highly parameterized neural networks trained on massive internet corpora, and overall, the computational, non-linear mechanisms underlying their predictions remain elusive. A second challenge stems from the way LLMs are evaluated, and, perhaps more importantly, how their performance is interpreted. It is now widely accepted in the NLP community that the conventional evaluation paradigm of static benchmarking is a rather misleading measure of models' true capabilities (McCoy et al., 2019; Zhang et al., 2022; Luo et al., 2024; Press et al., 2023; Mondorf and Plank, 2024). This critique is motivated by a series of paradoxical empirical findings. On the one hand, LLMs achieve state-of-the-art (SOTA) results, on many benchmarks(Devlin et al., 2019; Radford et al., 2019; OpenAI, 2024) even in tasks that target demanding phenomena, often associated with different forms of reasoning and the ability to draw inferences (Wei et al., 2023). Yet, the same models do not generalize effectively out-of-distribution conditions or real-world data (McCoy et al., 2019; Liu et al., 2023a), report big performance drops on challenge datasets (Nie et al., 2020a) and have been shown to replicate shallow statistical cues rather than generalize rules of logic (Zhang et al., 2022). These findings suggest that, despite the surge of enthusiasm surrounding LLMs, these systems often remain far from acquiring the high-level skills that their empirical success is often taken to imply.

In this context, there is a growing interest surrounding the reasoning abilities of foundation models. Although the concept of AI reasoning is still poorly defined (Yu et al., 2023), *deduction*, namely the ability to draw valid inferences from available information, is considered one of its central components, and a key capability that is theoretically useful for many downstream tasks such as dialogue modeling (Nie et al., 2020b) and summarization (Liu et al., 2023c). Natural Language Inference (NLI) is a

popular NLP task that is often treated as a benchmark for Natural Language Understanding (NLU). NLI formalizes the complex skill of logical inference, requiring models to determine whether a hypothesis logically follows from a given premise. Its appeal lies in the fact that models should, in theory, synthesize lower-level linguistic information, including semantic relations and syntactic structure, into higher-level semantic understanding McCoy et al. (2019). Nevertheless, much prior work has shown that NLI models tend to solve the task in unintended ways, instead of employing relevant linguistic features and principled inferential strategies akin to human reasoning and logical formalisms. To better understand these limitations, there has been increasing interest in diagnostic NLI datasets that isolate specific inferential phenomena under carefully controlled conditions(McCoy et al., 2019; Richardson et al., 2019).

One such phenomenon is *monotonicity reasoning*, a subtype of deductive inferences governed by systematic patterns comprised of lexical semantic relations and logical operators [1]. An example of monotonic inference is "*All dogs* barked" $\rightarrow$ "*All beagles* barked", where the lexical substitution *dog* $\sqsupset$ *beagle*, under the syntactic scope of the operator "All" yields an entailment relation between the premise and the hypothesis. Although this type of syllogisms have long been studied in Formal Semantics and successfully modeled by rule-based systems (van Benthem, 1986; MacCartney and Manning, 2009), learning the rule-like logic of monotonic reasoning implicitly from examples has proven challenging for the probabilistic paradigm of neural language models (Yanaka et al., 2019b; Geiger et al., 2020; Chen and Gao, 2023). Therefore, it serves as a useful test case for evaluating the reasoning strategies employed by NLI models and identifying their potential limitations.

1.1 Motivation and Research Focus

In this thesis, we investigate how two transformer-based models, GPT-2 and BERT-base, handle the phenomenon of Monotonic Reasoning. The motivation behind the methodology and experiments conducted in this work is overall driven by the following insights:

(a) Foundation models are not explicitly designed to reason. Yet, large-scale pretraining and their inherent ability to capture statistical patterns after being exposed to targeted data allow them to perform meaning-heavy tasks, such as that of inference, which in principle require complex linguistic and reasoning-like computations. However, the extent to which task-intended features are indeed employed by these systems remains an open question that much recent behavioral (McCoy et al., 2019; Yanaka et al., 2020) and representation-level (Geiger et al., 2020; Chen and Gao, 2022; Rozanova et al., 2024) studies have tried to address.

(b) It remains unclear which capabilities emerge at the pretraining phase and which are acquired through task-specific fine-tuning (Talmor et al., 2020).

(c) By definition, LLMs are statistical systems, designed to extract patterns from data. Consequently, a large body of work highlights that our expectations surrounding these models should be in line with the data presented to them in the

[1] A more detailed description of Monotonic Reasoning is provided in Chapter 2, Section 2.1.2.

fine-tuning stage, and how useful this information actually is for them to generalize to a given test distribution (Linzen, 2020; McCoy et al., 2023; Hupkes et al., 2023).

(d) NLI models often achieve high accuracy by exploiting spurious, non-generalizable correlations between features and labels(Poliak et al., 2018; Gururangan et al., 2018; McCoy et al., 2019). However, popular NLI benchmarks such as SNLI (Bowman et al., 2015) and MNLI (Williams et al., 2018) have been criticized for containing annotation artifacts and lacking the necessary linguistic control to diagnose such behaviors (Talman et al., 2021; Luo et al., 2022). Therefore, datasets that isolate specific linguistic phenomena within the broader task of inference are generally viewed as more effective methods for interpreting the strenghts and weaknesses of NLI models (McCoy et al., 2019; Yanaka et al., 2019a,b; Richardson et al., 2019; Geiger et al., 2020).

Given this background, we introduce a new diagnostic dataset, *MoRe* (Monotonic Reasoning), which contains premise-hypothesis pairs arising from the interaction between two specific linguistic components, namely hypernymy (e.g. cat $\sqsubset$ animal, flower $\sqsupset$ rose) and logical operators (e.g. *all, some, no*). To the best of our knowledge, MoRe is the first large-scale resource that targets this specific sub-case in this depth and linguistic control, and therefore constitutes one of the main contributions of this work. Given that NLI models struggle to acquire this type of reasoning in a robust and systematic manner, (Yanaka et al., 2019b; Geiger et al., 2020; Chen and Gao, 2023), we organize our dataset into three experimental splits designed to investigate which aspects of this phenomenon a model can capture, what linguistic information needs to be present during training, and under which conditions it can successfully generalize this knowledge on a test distribution.

Our methodology is organised into three main phases. First, we investigate whether GPT-2 contains sufficient information to handle general and monotonic NLI without being exposed to relevant data. To this end, we evaluate the model in a zero-and a few-shot prompting setting, as this is the "purest" setup for assessing pre-trained knowledge. Second, we examine whether information acquired from a general-purpose NLI benchmark, namely SNLI (Bowman et al., 2015) transfers to the monotonic inferences of MoRe. Finally, we conduct a series of finetuning experiments directly on our dataset. In particular, we introduce three experimental splits, where we carefully manipulate the linguistic features of hypernymy and logical operators. Our first two tests allow us to investigate if our models encode and rely on this linguistic information when they generalize across each training/test split. Our third split constitutes an extreme diagnostic setting, where we test the learnability of a strong shortcut signal. Since fine-tuned language models are expected to learn the statistical regularities that dominate a training distribution, we do not expect our models to overcome this biased signal. Instead, this experiment is designed to expose their tendency to capture and reproduce shortcut solutions from data, regardless of their relevance to the task logic.

1.2 Structure

In Chapter 2, we present the theoretical background of this thesis. We discuss the phenomenon of Monotonic Reasoning in Natural Language Inference (NLI) and review the main challenges that transformer-based models face in handling this type of reasoning,

focusing on evidence from previous studies on generalization, probing, and the problem of shortcut learning. In Chapter 3, we describe the design process behind our dataset. In particular, the chapter outlines all the steps required to collect our samples, including the design of a lexical taxonomy as well as the collection of hypernym-hyponym pairs and sentence templates. It also presents the dataset statistics and introduces the three diagnostic experimental splits that were used to train our models. Our experimental setting and methodology are presented in Chapter 4. The chapter begins by outlining the motivation behind our experiments, as well as describing how we trained GPT-2 and BERT-base for our In-Context-Learning and fine-tuning experiments. It then introduces our hypotheses and research questions, grounding them in the existing literature. In Chapter 5, we present the results of our experiments, as well as conduct an error analysis on our experimental splits. Finally, Chapter 6 contains a summary and reflection of our findings, addresses certain limitations of our approach, and provides suggestions for future research.

Chapter 2

Background and Related Work

2.1 Reasoning in Language Models

At the risk of stating the obvious, language is a notoriously complex object of study. Being the medium that allows us to accuse someone of *making a leap of logic*, warn against *jumping to conclusions*, or characterize a situation as one that *doesn't make sense*, language has played a central role in the study of Logic and Reasoning, two concepts that have been puzzling traditional disciplines for centuries.

The ability to draw inferences is a central aspect of human reasoning, which is broadly defined as the act of deriving conclusions from premises (Smith, 1989; Holyoak and Morrison, 2005; Hanna, 2006; Copi et al., 2011). When we as humans read that "Bruno is a dog" and "All animals need food to survive", we can infer with ease that "Bruno needs food to survive", as we know what individual components mean (dog, animals, food, survive) and stand for (e.g. animals and dogs as real-world entities) as well as understand their combined relation (dog $\subset$ animal) within the syntactic and logical context of a sentence. Probably one of the earliest attempts to represent language within a logical framework is Aristotle's study of syllogisms, which describes the conditions under which a conclusion can logically follow from a set of premises (Smith, 2022). As illustrated in the two following examples, this process may be carried out correctly, yielding valid inferences, or incorrectly, resulting in fallacies:

Valid Deduction	**Invalid Deduction**
(Aristotle, *Prior Analytics* I.1)	(Copi et al., 2011)
All humans are mortal.	All dogs are animals.
Socrates is a human.	All cats are animals.
Socrates is mortal.	**All cats are dogs.**

Aristotle's syllogism is often seen as a model case of *deduction*[1], while the process that allows us to move from given statements to a logical conclusion is called *inference* (Smith, 1989; Copi et al., 2011; Hurley, 2000). In the tradition of Formal Logic, the validity of an argument depends on its logical form rather than on its specific content.

[1]In logic, reasoning is typically divided into three types: (a) *deduction*, when the conclusion follows necessarily from the premises, as illustrated in Aristotle's syllogism; (b) *induction*, when the conclusion is probable based on observed evidence, e.g. Every yolk I have ever seen in yellow → All yolks are yellow; and (c) *abduction*, when we infer the most likely explanation for an observation, e.g. The ground is wet → It has probably rained). NLI benchmarks often contain all three types of reasoning.

Because of this property, arguments can be modeled through symbolic formulas that capture various patterns of inference:

	All philosophers are eccentric. Jack is a philosopher. $\therefore$ Jack is eccentric.	
All F are G n is F $\therefore n$ is G	All robins have red breasts. Tweety is a robin. $\therefore$ Tweety has a red breast.	*Smith* $(2004, pp.\ 5-6)$
	All existentialists write bosh. Sartre is an existentialist. $\therefore$ Sartre writes bosh.	

Formalizations as such were largely employed in the early stages of Artificial Intelligence, with the goal of developing systems that operate in a logically coherent way. Until the late 1980's, the field relied on symbolic systems, often referred to as *Good Old-Fashioned AI* (Haugeland, 1985). Within this paradigm, linguistic knowledge and reasoning were modeled through symbolic rules of Formal Logic, ontologies, and explicit knowledge representations. While such approaches offered a high degree of transparency and interpretability, they were viewed as inflexible, considering that a finite knowledge base could never encode enough information to generalize across different domains and deal with the complexity and variability of natural language (Luo et al., 2024).

Despite these apparent limitations, even these early symbolic systems started to give rise to the ambition of Strong AI, driven by the assumption that sufficiently sophisticated machines could approximate human-like reasoning. Opponents of this view criticized this vision for falling into scientific and conceptual paradoxes. A popular critique was expressed by Searle, who argued that although computers manipulate symbols well enough to solve a range of tasks, they do so without understanding their meaning so the way they operate is fundamentally different to human intelligence; *"all the computer has is more symbols* [...] *we can describe them as taking information in at one end, transforming it, and producing information as output"* but *"it is up to outside observers to interpret the input and output as information in the ordinary sense"* (Searle, 1980, p. 433).

2.1.1 Natural Language Inference (NLI)

Data-driven approaches enabled by advances in Machine Learning, and in the most recent years, the paradigm of Transformer-based Large Language Models (Vaswani et al., 2023) shifted the focus from the modeling of reasoning to the design of benchmarks, conventionally in natural language, that can help us investigate reasoning-like behaviours in language systems. In this thesis we focus on Natural Language Inference (NLI), a popular NLP task, often treated as a proxy assessment of Natural Language Understanding (NLU). In its most standard form, NLI is framed as a categorical classification over a pair of two statements, the *premise* [P] and the *hypothesis* [H]. An NLI model has to inspect the given sentences, infer their underlying logical relationship and classify it as (i) *entailment*, if the hypothesis logically follows from the premise, (ii)

contradiction, if the hypothesis is incompatible with the premise or (iii) *neutral*, if the premise neither follows from nor refutes the hypothesis.

[P] Annie is playing the guitar	[P] Joe is sleeping	[P] Two children are playing
[H] Annie is playing music	[H] Joe is running	[H] The children are siblings
↓	↓	↓
Entailment	**Contradiction**	**Neutral**

Large-scale datasets like SNLI (Bowman et al., 2015) and MNLI (Williams et al., 2018) have been widely used to train and diagnose NLI models on the ability draw logical inferences. In recent years, state-of-the-art models achieve notably high performance on these datasets, as well as in open open-ended inferential tasks like multi-hop question answering (Mavi et al., 2024) and proof generation (Tafjord et al., 2021). However, prior work on adversarial NLI has shown that many reasoning phenomena underlying such inferences do not emerge robustly in current models (e.g. McCoy et al. (2019)). One relatively underexplored case is Monotonicity, a reasoning pattern where neural models exhibit systematic weaknesses (Yanaka et al., 2019a,c; Richardson et al., 2019; Chen et al., 2020; Geiger et al., 2020).

2.1.2 Monotonic Reasoning

While it is probably impossible to provide a fully systematic account of how reasoning operates, given that language is "ambiguous at all structural levels" (Schubert, 2020) and meaning is to a large extent grounded in actual use and context (Wittgenstein, 1953; Firth, 1957; Grice, 1968; Halliday, 1976), a large body of work in Formal and Natural Logic has investigated types of arguments with highly consistent structure. For instance, MacCartney and Manning (2009) explain that many premise-hypothesis pairs rely on the interaction between lexical substitutions (e.g. synonymy: *couch* $\equiv$ *sofa*, hyponymy: *beagle* $\sqsubset$ *dog,* hypernymy: *dog* $\sqsubset$ *beagle*) and syntactic operators (e.g. [↑] some, many, at least n, [↓] no, few, hardly any), as in [P] *Some beagles* run to catch the ball [H] *Some dogs* run to catch the ball.

Monotonicity is an important property of many natural language inferences, that determines whether entailment is preserved or reversed when a lexical term is replaced by another with a broader or narrower meaning (van Benthem, 1986; MacCartney and Manning, 2009). As seen in Figure 2.1, monotonic contexts can be either upward or downward entailing. In upward-entailing contexts, operators such as *some* preserve entailment from 1(a) to 1(b) when the subset (*roses*) is replaced by the superset (*flowers*). In the second example, the quantifier *all* licenses a downward monotonic context, as in this case entailment emerges when the superset (*birds*) in 2a is substituted by the subset (*sparrows*) in 2(b). The type-of relation that characterizes *roses* $\sqsubset$ *flowers* and *birds* $\sqsupset$ *sparrows* is called *hypernymy* (Cruse, 1986; Lyons, 1977), where the more general term (i.e. flower, bird) is the *hypernym* and the more specific term (i.e. rose, sparrow) is the *hyponym*.

Monotonic NLI is generally viewed as a compositional reasoning task, in the sense that it comprises highly systematic semantic and syntactic cues that a model must attend to and generalize over in order to arrive at a correct inference label (Geiger et al., 2020). This inherent systematicity prompted earlier studies to model this type of reasoning in formal or natural-logic frameworks (MacCartney and Manning, 2009;

Upward Monotonicity

(1) (a) Some *roses* bloomed early
(b) Some *flowers* bloomed early

Downward Monotonicity

(2) (a) All *birds* slept on the tree
(b) All *sparrows* slept on the tree



Figure 2.1: Monotonicity-driven entailment patterns.

Abzianidze, 2015; Mineshima et al., 2015) that underpinned rule-based reasoning systems. More recent work on data-driven NLI has focused on whether neural language models can implicitly approximate and generalize such reasoning patterns by being trained and tested on dedicated datasets. The scarcity of monotonic inference instances in general NLI benchmarks created the need for datasets that explicitly target Monotonicity in NLI, popular examples being HELP (Yanaka et al., 2019a), Monotonicity Entailment Dataset (MED) (Yanaka et al., 2019b), Semantic Fragments (Richardson et al., 2019) and Monotonicity Entailment Dataset (MoNLI) (Geiger et al., 2020) which we describe in more detail in the next section.

2.1.3 Review of Monotonic NLI Datasets

HELP

HELP (Yanaka et al., 2019a) is one of the largest synthetic Monotonicity NLI datasets (36k premise–hypothesis pairs) annotated as entailment/neutral. The dataset comprises sentences derived from the Parallel Meaning Bank, a multi-genre corpus, and uses WordNet as a source of lexical substitutions. HELP targets a range of monotonicity cases, including upward, downward and non-monotone inferences, as well as a range of logical operators, offering a broad coverage of inferential phenomena. Despite its overall significance, when we manually inspected HELP, we found that a considerable number of examples are linguistically ill-formed (presented in Appendix A, Table A.1).

MED

Monotonicity Entailment Dataset (MED) (Yanaka et al., 2019b) contains 5,382k premise hypothesis pairs, covering upward, downward and non-monotone cases, again labeled as entailment and non-entailment. The majority of entailments in MED arise through ellipsis (e.g. No delegate finished the report on time vs. No delegate finished the report), modifier deletion (e.g. Many delegates obtained interesting results vs. Many delegates obtained results) or coordination (e.g. He thinks nothing of poverty vs. He thinks nothing of poverty, nor about wealth).

Semantic Fragments

Semantic Fragments (Richardson et al., 2019) is a set of challenge datasets designed to probe models' sensitivity to a range of semantic phenomena, including monotonicity. The monotonic subset of Semantic Fragments comprises 10k examples, that follow a three-way label classification (entailment/neutral/ contradiction), thematically focusing on sentences from the animal domain. This dataset also includes a range of phenomena that interact with monotonicity, like for instance, embedded clauses of various syntactic depths (e.g. No *animals* who some beagle stared at *touched* a brown cat / No *bats* who some beagle stared at *did not touch* a brown cat).

MoNLI

Monotonicity Natural Language Inference (MoNLI) (Geiger et al., 2020) is a small-scale challenge dataset (2.700 instances) that targets downward monotonic reasoning arising from hypernymy under negation scope, constructed by applying hypernym–hyponym substitutions to existing SNLI sentence templates (e.g. The man is *not listening* to *music* vs. The man is *not listening* to *jazz*).

2.2 Transformers

The current state-of-the-art (SOTA) in the study of reasoning phenomena focuses on models such as BERT (Devlin et al., 2019) and GPT (Radford et al., 2019). At their very core, transformer-based LLMs are deep neural networks that rely on the *attention* mechanism, to learn the probability distribution of words present in massive amounts of training data. This large-scale unsupervised pre-training helps these models acquire latent forms of linguistic knowledge, something that allows them to specialize in various downstream tasks and benchmarks through training methods such fine-tuning and in-context learning (Brown et al., 2020; Liu et al., 2019; Wang et al., 2019a). One of the main assets of this new generation of models is their ability to capture intricate contextual relationships over long-range sequences of text, something that was viewed as a bottleneck in their predecessors, such as RNNs or LSTMs.

Evidence suggests that the complexity of information that models store tends to rise in a bottom-up way, starting from the encoding of surface-level features in the lower layers, progressing towards higher-level abstractions and semantic information in the upper layers (Tenney et al., 2019; Jawahar et al., 2019; de Vries et al., 2020; Rogers et al., 2020). However, LLM's decision-making is opaque and non-linear, which makes it difficult to determine in absolute terms what kind of information they rely on when they make a prediction. Probably one of the most intriguing open questions in NLP is whether in the course of this opaque learning process, models also acquire representations that align with linguistically meaningful structures (Yanaka et al., 2019a; Ravichander and et al., 2020; Olah et al., 2020; Richardson et al., 2019; Wang et al., 2022; Donatelli and Koller, 2023), and linguistic theory in general. To date, researchers remain divided on whether the empirical gains observed in LLMs arise from their improved modeling of meaning or from an enhanced ability to memorize co-occurrence patterns in text corpora (Pavlick, 2022).

2.2.1 BERT and GPT-2

Many contemporary models relied on the original Transformer architecture, but adopted different designs and training objectives that made each stand out in different aspects of language processing. BERT (Devlin et al., 2019) is an autoencoding model trained on on Masked Language Modeling (MLM) where some input words are randomly hidden and must be recovered from context, and Next Sentence Prediction (NSP), where given a sentence A and B the model decides whether B follows logically from A. BERT encodes input in deep bidirectional word representations given the entire left and right context, a property that makes it suitable for tasks such as Natural Language Inference or Sentiment Analysis, that require full contextual understanding to be solved effectively. Autoregressive models like GPT (Radford et al., 2019) are trained on a Causal Language Modeling (CLM) objective, meaning that they learn to predict the next token sequentially, relying solely on the preceding left words. GPT's unidirectionality allows it to produce coherent text, a skill required in open-ended tasks like summarization, translation and question answering, but it may pose limitations for certain classification problems that require a holistic understanding of context.

2.2.2 Fine-tuning and the Problem of Generalization

In NLP, generalization is viewed as a key indication that a system has learnt the underlying structure of a task, typically understood as the ability to transfer knowledge to new, unseen instances (Hupkes et al., 2023). A common framework for studying generalization is the distinction between in-distribution (ID) and out-of-distribution (OOD) evaluations. In standard NLU benchmarks, training and test data are assumed to be independent and identically distributed (i.i.d.), so models like BERT are typically fine-tuned on the training distribution of a dataset and evaluated on a test partition that shares the same underlying patterns, statistical properties, and linguistic cues. As discussed so far, this is the setting where current SOTA models perform exceptionally well. For example, Table 2.1 shows that finetuned BERT and GPT-2 achieve strong ID performance on SNLI (Bowman et al., 2015) and MNLI (Williams et al., 2018).

Dataset	BERT-base	GPT-2	Source
SNLI	90.4	87.0[1]	Talman and Chatzikyriakidis (2019)
MNLI	∼85.0	82.0[2]	Devlin et al. (2019); Talman and Chatzikyriakidis (2019)

Table 2.1: BERT-base and GPT-2 on general-purpose NLI benchmarks (in-distribution).

Nevertheless, strong performance on a benchmark does not imply that a model has learned the underlying task itself, so standalone in-distribution accuracy is viewed as a rather misleading measure of models' reasoning behaviour (McCoy et al., 2019; Zhang et al., 2022; Luo et al., 2024; Press et al., 2023; Mondorf and Plank, 2024). In the case of Monotonicity, empirical findings show that when models are trained on broad-coverage NLI benhcmarks they perform below chance on monotonic datasets

[1]To the best of our knowledge there is no finetuned evaluation of GPT-2 on SNLI or MNLI. The only available evidence comes from publicly released models on Hugging Face, so we cite these results with caution `https://huggingface.co/varun-v-rao/gpt2-snli-model2`

[2]`https://huggingface.co/tanganke/gpt2_mnli/blob/main/results.csv`

(Geiger et al., 2020; Rozanova et al., 2021, 2022; Chen and Gao, 2023; Wijnholds, 2023). Rozanova et al. (2021) show that performance improves only when models are exposed to additional monotonicity-specific supervision during training. For example, as we described in Section 2.1.3, MoNLI (Geiger et al., 2020) targets downward monotonic reasoning arising from hypernymy under negation scope (e.g. The boy is *not eating* a *fruit* vs. The boy is *not eating* an *apple*). Table 2.2 shows that BERT performs slightly above chance when trained solely on SNLI, and improves only marginally when another monotonicity dataset, HELP (Yanaka et al., 2019a), is added during training. However, performance increases significantly only when BERT is exposed to held-out examples from MoNLI itself.

Model	**Additional Data**	**MoNLI**
bert-base-uncased-snli	-	46%
bert-base-uncased-snli	+ HELP	55%
bert-base-uncased-snli	+ MoNLI	90%

Table 2.2: BERT's performance on MoNLI under data augmentation. Results obtained by Rozanova et al. (2021) and Geiger et al. (2020).

Previous NLI studies have expressed the idea that if a model has actually acquired the ability to approximate the skill of logical inference, it should, at least in theory, be able to generalize across different datasets that represent the same phenomenon with slight variations (Talman and Chatzikyriakidis, 2019). From this perspective, generalization problems give rise to the hypothesis that NLI models tend to over-fit to training data without capturing task-relevant abstractions. In line with this view, Miralles-González et al. (2025) explain that data-centric fine-tuning is unlikey to result in robust generalization, unless the necessary representations have already emerged during the model's pre-training phase. This intuition has prompted researchers to diverge from standard end-to-end benchmark evaluations and adopt more interpretable evaluation methods to gain clearer insights into the aforementioned shortcomings.

2.3 Evidence from Monotonic NLI Literature

Although there is still a plethora of open questions in the study of neural Monotonic Reasoning, as a relatively small body of work has investigated the phenomenon using Transformer-based models, the majority of existing studies conclude that monotonicity reasoning does not emerge robustly in SOTA NLI models like BERT and GPT (Richardson et al., 2019; Yanaka et al., 2019b; Chen and Gao, 2023). We notice that prior work on Monotonicity NLI can broadly be divided into two research directions. One line of work employs probing methods to inspect the internal representations of neural models (Richardson et al., 2019; Goodwin et al., 2020; Rozanova et al., 2021), examining whether linguistic properties relevant to monotonic inference are encoded within hidden representations. In practice, this often involves probing for the semantic relation of hypernymy (dog $\supset$ beagle, peach $\subset$ fruit) or for the representations associated with logical operators (like *all*, *some*, *no*) and negation (Geiger et al., 2020). A second body of work frames Monotonicity NLI as a problem of systematic generalization, approaching this type of reasoning as an adversarial test case for studying

whether models can generalize on unseen reasoning patterns. In the next two sections, we present some of the main findings from these two setups, followed by a discussion on the problem of shortcut learning, which is a widely-discussed hypothesis behind many undesired behaviours observed in NLI models.

2.3.1 Systematic Generalization

Systematicity and *compositionality* have long been viewed as defining elements of human language intelligence and understanding (Goodman et al., 2008). In recent years, these notions have also emerged as evaluation criteria in NLP, particularly in assessing whether neural models generalize beyond their training distribution in a structured and principled way (Lake and Baroni, 2018; Nie et al., 2018; Hupkes et al., 2020). While there is a broad agreement that neural networks struggle to exhibit fully compositional and systematic generalization (Pavlick, 2022; Goodwin et al., 2020; Vegner et al., 2025), there is less consensus on how these concepts should be defined and tested in the context of LLMs (Hupkes et al., 2020).

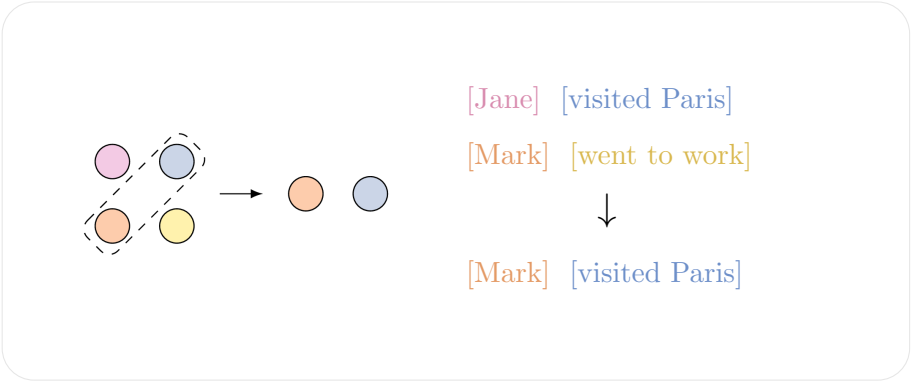


Figure 2.3: Schematic Illustration of Systematicity in Natural Language. Adapted from Hupkes et al. (2020).

The linguistic principle of compositionality states that "the meaning of a complex expression is determined by its structure and the meanings of its constituents". (Pickel and Szabó, 2025). Closely related is systematicity, which concerns the ability to generalize learned knowledge across structurally related contexts (Vegner et al., 2025). A classic example is provided by Fodor and Pylyshyn (1988), who argue that humans and a potentially robust system that understands a statement like "John loves Mary" should also be able to understand "Mary loves John". For instance, Lake and Baroni (2018) introduced SCAN, a benchmark for compositional generalization, where models are trained on commands such as "walk twice" = WALK WALK and "run twice" = RUN RUN, and are expected to infer that "jump twice" also translates to JUMP JUMP, abstracting the general rule associated with the operator "twice".

Prior evidence from Monotonic NLI shows that while models may exhibit some degree of systematicity, this behaviour is often restricted to superficial patterns. For instance, Yanaka et al. (2020) showed that BERT can generalize to unseen combinations of lexical relations and quantifiers. In their setup, models were trained on examples such as [Some dogs in the park ran], [Several dogs run], [No puppy ran] and tested on unknown combinations such as [Several dogs in the park run] and [No dog in the park run]. The authors found that BERT failed to generalize when syntactic depth was

introduced (e.g. [Some dogs that chased cats that scared birds barked]), suggesting that their model relied on surface-level cues.

Yanaka et al. (2019b) showed that BERT's performance on monotonicity reasoning depends strongly on the proportion of upward- and downward-entailing examples in the training data. Rather than learning abstract monotonicity patterns that could help the model generalize consistently across inference directions, BERT improved on the direction that dominated the training distribution, while performance on the opposite direction deteriorated. These findings already indicate that models likely adopt shortcut strategies inherent to each direction of inference, rather than handle monotonicity compositionally. This interpretation is further supported by a general observation that LLMs perform substantially better on upward than downward-entailing inferences (Wang et al., 2019b; Rozanova et al., 2021; Chen and Gao, 2023; Rozanova et al., 2024), and struggle considerably when negation is involved (Geiger et al., 2020).

Geiger et al. (2020) examined whether models can learn the interaction between lexical entailment and negation in MoNLI. Their findings indicate that SNLI-trained BERT captures lexical relations in isolation (sneakers $\subset$ shoes), achieving 94% accuracy on cases like [P] The girl was wearing *sneakers* [H] The girl was wearing *shoes* ($\rightarrow$ entailment). On the other hand, their model could not understand that negation flips inference, failing almost completely when tested on examples when the same substitutions appeared under the scope "not" (as in [P] The girl was *not wearing sneakers* [H] The girl was *not wearing shoes* $\rightarrow$ neutral). As a systematic evaluation test, the authors tested if BERT can learn the rule "negation flips the label" by testing BERT on lexical substitutions that were not seen during training (e.g. trained on [sneakers$\subset$shoes], [animal$\supset$dog], and tested on [apples$\subset$ fruit], [flowers$\supset$roses]), a setting where BERT performed really well ($\sim$90%).

2.3.2 Probing

Another line of work on Monotonic reasoning employs probing methods. In simple terms, probing involves freezing a model specialized on a downstream task, extracting its hidden representations, and training an auxiliary classifier on them to predict a linguistic property Belinkov (2021). Evidence suggests that BERT and other transformers encode hypernymy-related information to a considerable extent, yet we notice the opposite tendency for the representations of quantifiers, which appear to be weak and inconsistent (Nie et al., 2018; Richardson et al., 2019; Rozanova et al., 2021; Goodwin et al., 2020; Chen and Gao, 2023; Rozanova et al., 2024). For instance, (Rozanova et al., 2021) showed that BERT fails to represent negation and quantifiers, treating downward-monotone contexts as if they were upward-monotone. Related findings show that models often ignore the semantic contribution of quantifiers such as *all*, *some*, *no*, and *most* (Yanaka et al., 2019b,c; Richardson et al., 2019), a tendency that is again far from what we could characterize as a compositional-like approach to inference. It should be noted that positive probing results do not necessarily imply that the model actually uses this information during inference (Glockner et al., 2018; Belinkov, 2021), given that probes are linear, while LLMs handle information in non-linear ways.

One possible hypothesis is that when handling this type of reasoning, neural models may exhibit a behaviour characterized as a compositionality gap. In particular, Press et al. (2023) suggest that there might be a discrepancy between a model's ability to handle individual sub-problems and its ability to compose them into a correct overall solution. For instance, Ettinger (2020) showed that even though BERT's embeddings

encode hierarchical meaning, the model was not able to use this information in cloze-style tests consistently. In particular, the authors found that BERT correctly predicted [bird] for the masked prompt "A robin is a [MASK]" but provided the same answer in "A robin is *not* a [MASK]". This finding suggests that BERT likely encodes that *robin* and *bird* are related, but it cannot integrate this information compositionaly in the context of a sentence. On the other hand, certain studies raise doubts that LLMs represent hypernymy as a structured relationship in the first place, leaning towards the explanation that models may simply memorize taxonomic information from pre-training (Ravichander et al., 2020; Hanna and Mareček, 2021).

Following Zhang et al. (2022), many logical problems are solved through an underlying "reasoning function" that represents the desired solution to the task. As discussed so far, we find little supporting evidence that models genuinely capture the compositional argument structures required for such reasoning. In the next section, we explain that the discussion surrounding shortcut learning is probably the strongest argument against the hypothesis that LLMs reason in a systematic manner. At least as far as NLI is concerned, it is well established from a variety of studies that NLI models do not use task-informative features during inference.

2.3.3 Shortcut Learning in NLI

A widely-discussed hypothesis behind LLMs' generalization weaknesses and vulnerability to adversarial attacks is their tendency to adopt spurious decision rules, a behaviour known as *shortcut learning*. In simple terms, the very process of training a model to learn from a dataset exposes it to statistical patterns inherent to its examples. Following Ilyas et al. (2019), this problem space contains three types of features that a model can capture: (a) useless features (b) non-robust features and (c) robust features. Supervised fine-tuning likely drives NLI models towards adopting features (a) and (b), the so-called shortcuts. On the other hand, robust features represent decision rules that capture informative reasoning abstractions, which could, in theory, help a model generalize to all relevant out-of-distribution evaluations. Yet, Geirhos et al. (2020) describes shortcut learning as a computational instantiation of the *Principle of Least Effort* (Zipf, 1949), as even though multiple predictive strategies are available, neural models tend to rely on those that are statistically stronger and thus easier to learn and apply.

Song et al. (2024) distinguish between intrinsic and extrinsic shortcuts. During pre-training, LLMs are exposed to large-scale corpora in order to learn broad statistical regularities of language. At this phase, they may already form unintended correlations that can later manifest as shortcut behaviors. For instance, McKenna et al. (2023) report on a memorization bias in NLI, as they found that hypotheses expressing common factual statements (e.g. Whales are mammals) pushed many model families (LLaMA, GPT-3.5, and PaLM) towards predicting entailment, even though factuality was not an informative feature for the inference relation. Conversely, extrinsic shortcuts emerge during task adaptation. McCoy et al. (2019) describe three popular non-robust heuristics that NLI models exploit from training data; (a) *subsequence*, when the hypothesis is contained within the premise, *constituency*, when the hypothesis forms a syntactic part of the premise and (c) *lexical overlap*, when the premise and hypothesis share many overlapping words. A striking case is the hypothesis-only bias (Poliak et al., 2018; Glockner et al., 2018; Tsuchiya, 2018; McKenna et al., 2023) as NLI models achieve surprisingly high accuracy without even seeing the premise. Gururangan et al. (2018) showed that negation words are highly predictive of the contradiction label, including

cases where negation does not logically imply contradiction.

Du et al. (2023a) mention that pretraining objectives and model size should be viewed as contributing factors. They argue that smaller models are not only less capable in terms of generalization, but also more prone to exploiting spurious correlations, providing evidence from model distillation literature (Jin et al., 2020; Du et al., 2023b). However, other studies show that the problem of shortcut learning is not alleviated through scaling, leaving the impact of model size an open question (McKenzie et al., 2024). For instance, Tang et al. (2023) found that GPT2-large is significantly more prone to exploiting shortcuts in prompts than GPT2-base for sentiment classification. Yuan et al. (2024) suggest that larger models rely more heavily on NLI shortcuts under zero-shot and few-shot prompting within the LLaMA 2 family. Chen and Gao (2023) further demonstrate that larger NLI models do not have a generalization advantage in monotonic inference. Overall, these findings suggest that NLI models often adopt decision rules that only work under limited conditions, urging caution in how we interpret a benchmark score.

Chapter 3

Dataset Creation

In this chapter, we describe the design process behind our diagnostic NLI dataset on Monotonicity Reasoning (MoRe)[1]. MoRe is designed to evaluate the ability of an NLI model to capture and generalize over carefully controlled patterns of monotonic reasoning, arising from the compositional interaction between the lexical relationship of hypernymy and logical operators. Although this is one of the most fundamental monotonic reasoning patterns described in the literature (van Benthem, 1986; MacCartney and Manning, 2009), only a limited number of existing resources have focused on this specific type of inference (for instance Geiger et al. (2020)). MoRe contains 927,360 premise-hypothesis pairs and, to the best of our knowledge, constitutes the largest-scale resource targeting this phenomenon.

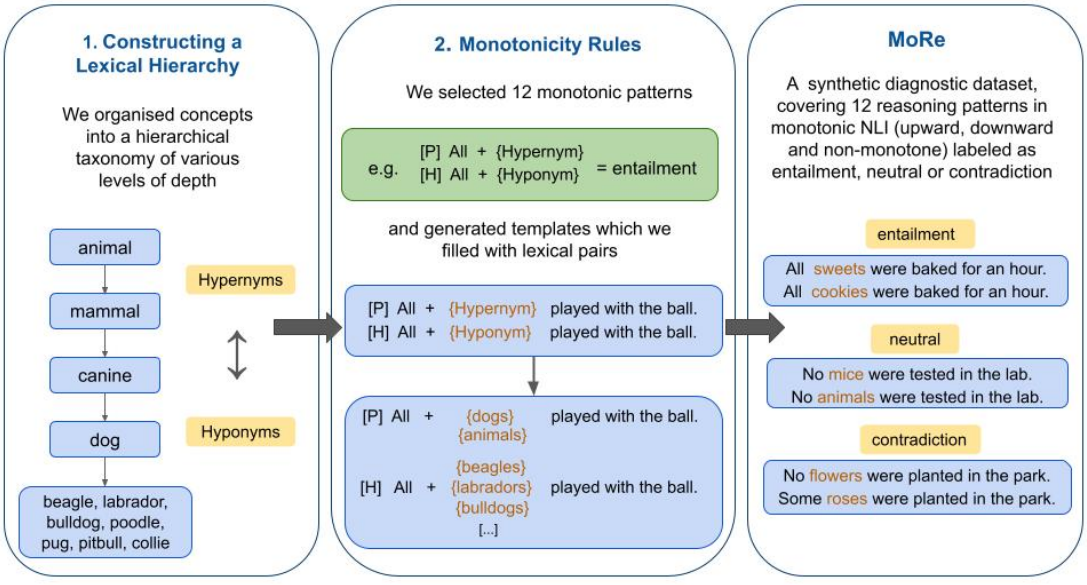


Figure 3.1: Schematic Overview our Methodology.

As illustrated in Figure 3.1 our methodology can be summarized in three main phases. First we selected 12 monotonic inference rules (see Section 3.1) that we wanted to sample. To generate the desired instances, we designed a lexical taxonomy and inserted the collected hypernyms and hyponyms in premise-hypothesis templates that

[1]The code used to create MoRe is available at this repository.

reflect the logic of our selected rules (see Section 3.3). A more detailed description of our methodology is presented in the following sections.

3.1 Selecting Monotonic Inference Patterns

In the previous chapter, we explained that monotonic inferences manifest as systematic reasoning patterns governed by predictable combinations between lexical relations and logical operators. Building on prior work on Monotonicity in Natural Logic (van Benthem, 1986; MacCartney and Manning, 2009), we selected 12 monotonic inference rules as the basis for sampling our instances. Table 3.1 summarizes the rules underlying our dataset grouped by quantifier type (e.g. *all, some, no*) and monotonic direction (i.e. upward, downward and non-monotonic).

Rule	Dir	Premise	Hypothesis	Label	Explanation
I. Quantifier 'All' (Universal Quantifier)					
R1	⇓	All mammals searched for food.	All horses searched for food.	Entail.	If Horse $\subset$ Mammal, then $\forall x(Mammal(x) \rightarrow P(x)) \models \forall x(Horse(x) \rightarrow P(x))$. Under "all," moving from a general category to a specific one preserves entailment.
R2	⇔	All horses searched for food.	All mammals searched for food.	Neutr.	If Horse $\subset$ Mammal, then $\forall x(Horse(x) \rightarrow P(x)) \not\models \forall x(Mammal(x) \rightarrow P(x))$. Expanding from the subset to superset breaks entailment, creating a non-monotone context.
II. Quantifier 'Some' (Existential Quantifier)					
R3	⇑	Some poodles ran to catch the ball.	Some dogs ran to catch the ball.	Entail.	If Poodle $\subset$ Dog, then $\exists x(Poodle(x) \wedge P(x)) \models \exists x(Dog(x) \wedge P(x))$. "Some" creates an upward-monotone context, where generalization from subset to superset is valid.
R4	⇔	Some dogs ran to catch the ball.	Some poodles ran to catch the ball.	Neutr.	If Poodle $\subset$ Dog, then $\exists x(Dog(x) \wedge P(x)) \not\models \exists x(Poodle(x) \wedge P(x))$. Narrowing the domain under "some" breaks entailment, yielding a neutral inference relation.
III. Quantifier 'No' (Negative Quantifier)					

Rule	Dir	Premise	Hypothesis	Label	Explanation
R5	⇓	No shops were open in the town.	No bakeries were open in the town.	Entail.	If Bakery $\subset$ Shop, then $\neg\exists x(Shop(x) \wedge P(x)) \models \neg\exists x(Bakery(x) \wedge P(x))$. The quantifier "no" establishes a downward-monotone environment, where falsity for a superordinate category entails the falsity for its subset.
R6	⇔	No bakeries were open in the town.	No shops were open in the town.	Neutr.	If Bakery $\subset$ Shop, then $\neg\exists x(Bakery(x) \wedge P(x)) \not\models \neg\exists x(Shop(x) \wedge P(x))$. Negating over a subset does not entail negation for its superset.
IV. No Operator (Eventive Contexts)					
R7	⇓	Microbes grew inside the test tubes.	Streptococcus grew inside the test tubes.	Ent.	If Streptococcus $\subset$ Microbe, then P(Microbe) $\models$ P(Streptococcus) A property stated for a superset also holds true for the subset.[2]
R8	⇔	Streptococcus grew inside the test tubes.	Microbes grew inside the test tubes.	Neutr.	If Streptococcus $\subset$ Microbe, then $P(Streptococcus) \not\models P(Microbe)$. A property stated for a subset does not generalize to the superset.
V. Reversal and Contradiction Rules					
R9	⇓	All birds flew across the valley.	No sparrows flew across the valley.	Contr.	If Sparrow $\subset$ Bird, then $\forall x(Bird(x) \rightarrow P(x)) \wedge \neg\exists x(Sparrow(x) \wedge P(x))$ cannot both hold. The polarity shift between "all" and "no" introduces a contradiction.
R10	⇑	No birds flew across the valley.	All sparrows flew across the valley.	Contr.	If Sparrow $\subset$ Bird, then $\neg\exists x(Bird(x) \wedge P(x)) \wedge \forall x(Sparrow(x) \rightarrow P(x))$ is inconsistent. Reversing "no" to "all" changes polarity, yielding a contradiction.

[2] In formal semantics, sentences without a quantifier do not create a monotonic environment, and thus elicit a neutral label. We however marked R7 as entailment, parallel to the quantified cases.

Rule	Dir	Premise	Hypothesis	Label	Explanation
R11	⇑	No shrubs were planted in the park.	Some roses were planted in the park.	Contr.	If Rose $\subset$ Shrub, then $\neg\exists x(Shrub(x) \wedge P(x)) \wedge \exists x(Rose(x) \wedge P(x))$ cannot both be true. A negative quantifier conflicts with an existential assertion over its subset.
R12	⇓	Some roses were planted in the park.	No shrubs were planted in the park.	Contr.	If Rose $\subset$ Shrub, then $\exists x(Rose(x) \wedge P(x)) \wedge \neg\exists x(Shrub(x) \wedge P(x))$ yields contradiction. An existential affirmation over a subset cannot coexist with a universal negation over the superset.

Table 3.1 The table groups inference patterns by quantifier type and polarity behavior, showing how substitution between hypernyms and hyponyms elicits entailment, contradiction or neutral inference contexts. Purple terms indicate hypernyms and brown terms indicate hyponyms. The "Dir." column denotes monotonicity direction: ⇑ (upward entailment), ⇓ (downward entailment), and ⇔ (neutral or non-monotonic). All examples are retrieved from MoRe.

Symbol	Description
$\forall x$	Universal quantifier (*for all x*)
$\exists x$	Existential quantifier (*there exists x*)
$P(x)$	Predicate describing a property or event (*x performed P*)
$\subset$	Hyponymic inclusion relation (*subset of*)
$\models$	Logical entailment (*entails*)
$\not\models$	Lack of entailment / neutral inference
$\neg$	Negation (*not*)
$\wedge$	Logical conjunction (*and*)
$\Rightarrow$	Conditional implication (*if ... then ...*)
⇑ ()	Upward-monotone context (entailment preserved when generalizing)
⇓ ()	Downward-monotone context (entailment preserved when restricting)
⇔ ()	Non-monotone or neutral context (no entailment relation)

Table 3.2: Glossary of the symbolic notations used in Table 3.1

As seen in Table 3.1, MoRe contains (i) downward-entailing (R1, R5), (ii) upward-entailing (R3), (iii) non-monotone/neutral (R2, R4, R6, R8) and (iv) contradiction cases (R9–R12). One of our design goals was to include all three possible monotonicity directions and labels, a property that existing datasets lack. The most common practice that previous resources follow is a binary classification scheme between entailment vs. non-entailment (Yanaka et al., 2019a,b) or entailment vs. neutral (Geiger et al., 2020), while none so far contains all three directions (upward, downward, non-monotonic) and labels (entailment, neutral, contradiction). Given that each direction has been

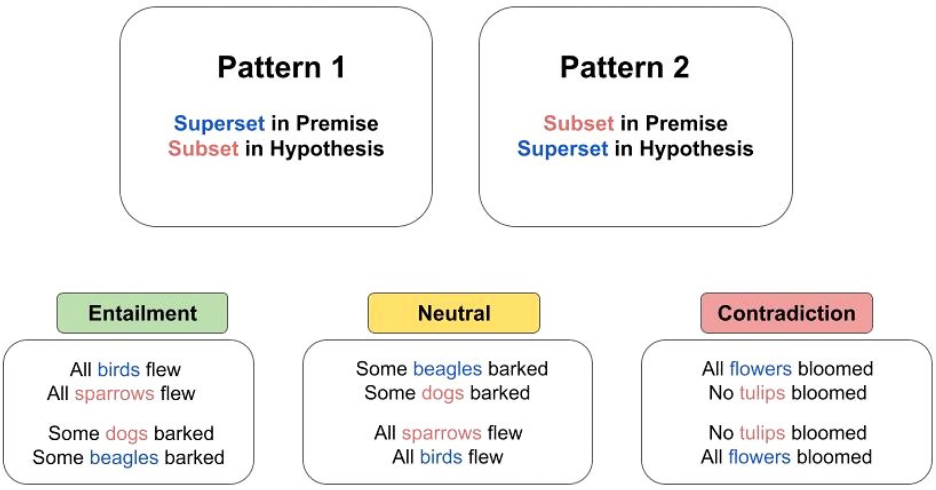


Figure 3.2: Reasoning Patterns of MoRe.

shown to pose different levels of difficulty for language models (Wang et al. (2019a); Rozanova et al. (2021); Geiger et al. (2020), we hope that the broader coverage of MoRe will provide clearer insights into the specific conditions where Monotonic NLI models succeed or fail. The selected inference patterns require a model to consider the interaction between lexical relations and the syntactic context in which they are embedded. As seen in Figure 3.2, our design exploits a core property of monotonicity, namely symmetry. This means that any monotonic pair (e.g. [P] All *birds* flew [H] All *sparrows* flew = entailment) can be inverted to yield a distinct, complementary pair if we reverse the position of the hypernym and the hyponym (i.e. [P] All *sparrows* flew [H] All *birds* flew = neutral).

3.2 Designing a Lexical Taxonomy

To populate the 12 reasoning patterns, we first had to generate hypernym-hyponym pairs that would subsequently be inserted in sentence templates. To ensure that this process is systematic and linguistically informed, we designed a lexical taxonomy[3] to organize concepts in a hierarchical way. To this end, three main steps were required:

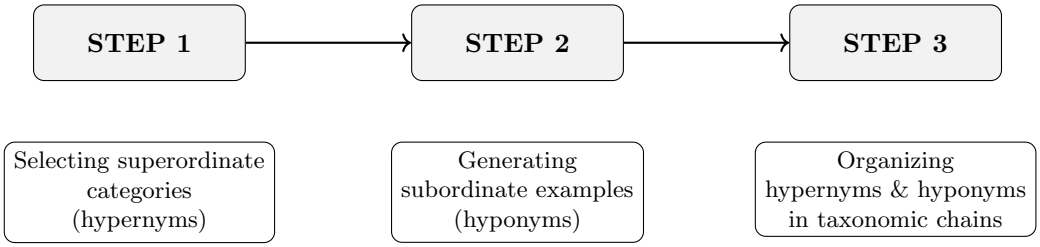


Figure 3.3: Overview of the Lexical Taxonomy Construction Process

[3]An interactive version of our taxonomy is available in this link.

3.2.1 Collecting Hypernyms and Hyponyms

The semantic relationship of hypernymy allows concepts to be placed in hierarchical relationships of various depths, starting from broader categories (the hypernyms) which function as the supersets of their members (the hyponyms). We collected hypernyms (levels 1 - 3) manually, consulting general-purpose sources like Wikipedia, particularly in the case of biological groups (e.g. *mammals* → *animals* → *vertebrates*) which follow strictly factual taxonomic ranks.

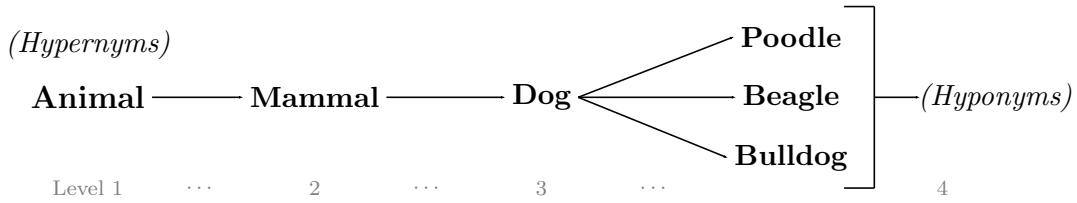


Figure 3.4: Example of a hierarchical hypernym–hyponym structure

To collect hyponyms (level 4), we concluded that the most scalable and efficient method was to prompt ChatGPT (OpenAI, 2025), following the paradigm of instruction-based prompting (Brown et al., 2020; Ouyang et al., 2022). In this context, the LLM is guided through natural-language instructions that specify the task as well as the format of the desired output. We opted for a zero-shot setting, meaning that no examples were included in the prompt beforehand. Below we provide our prompt as well as a sample output:

Initial Prompt

```
Provide 40 representative examples of {category}. Only include single-word
nouns and return the output as a Python list.
```

We encountered two main types of errors during this phase: (*i*) merged compounds such as *goldenretriever* and (*ii*) truncated compounds such as *german* or *grey*:

Sample ChatGPT Output for *dogs*

```
dogs = ["beagle", "bulldog", "poodle", "retriever", "labrador", "terrier",
"spaniel", "mastiff","pointer", "setter", "goldenretriever", "collie",
"shepherd", "greyhound", "whippet","chihuahua", "pitbull", "pug",
"dalmatian", "rottweiler", "grey","doberman", "husky","malamute", "boxer",
"mountaindog", "ridgeback", "dachshund", "akita","samoyed", "pinscher",
"beardedcollie", "corgi", "newfoundland", "borzoi","weimaraner", "saluki",
"basenji", "briard", "springerspaniel","german",komondor","eskimo",
"spaniel", "airedale", "malinois", "wolfhound"]
```

These issues made it necessary to refine our prompt by imposing stricter lexical constraints:

Refined Prompt

```
Provide 40 representative examples of {category}. Each item must be a
single-word noun.  Do not include any compound or multi-word expressions.
If an instance is originally a compound term, don't merge its parts,
simply exclude it.  It is also acceptable if this results in slightly
fewer than 40 items.  Return the output as a Python list of strings.
```

We repeated the refined prompt 44 times (once for each category) and organized our samples in a Python file, which was later used during the rule-generation phase.

3.2.2 Taxonomy Statistics

As illustrated in Table 3.3 we collected a taxonomic database consisted of 44 broad semantic categories and 2,862 hypernym-hyponym pairs, distributed across two general domains:

1. **Animate Concrete Categories**: animals and living entities
2. **Inanimate Concrete Categories**: non-living entities

Category	Hypn.	Pairs	Samples	Hypernym Chains
I. Animate (Concrete Categories)				
Mammals	51	153	dog, cat	mammals → animals → vertebrates
Dogs	55	224	akita, husky	dogs → canines → mammals → animals
Primates	25	78	monkey, gorilla	primates → mammals → animals
Rodents	17	63	mouse, squirrel	rodents → mammals → animals
Felines	14	42	cat, lion, tiger	felines → mammals → animals
Ungulates	27	120	horse, cow, pig	ungulates → mammals → animals
Marsupials	15	51	kangaroo, koala	marsupials → mammals → animals
Cetaceans	7	21	whale, dolphin	cetaceans → mammals → animals
Birds	40	123	sparrow, robin	birds → animals → vertebrates
Fish	62	195	shark, herring	fish → animals → vertebrates
Insects	31	105	ant, bee, beetle	insects → arthropods → invertebrates
Reptiles	21	63	snake, turtle	reptiles → animals → vertebrates
Amphibians	11	33	frog, toad	amphibians → animals → vertebrates
Arachnids	12	42	spider, scorpion	arachnids → arthropods → invertebrates
Mollusks	14	42	snail, oyster	mollusks → invertebrates → animals
Crustaceans	20	60	crab, lobster	crustaceans → arthropods
Dinosaurs	40	40	raptor	dinosaurs
Microbes	21	50	salmonella	microbes → microorganisms
Bacteria	24	48	mycoplasma	bacteria → microbes
Protozoa	25	50	toxoplasma	protozoa → microorganisms
Viruses	40	120	coronavirus	viruses → pathogens → microorganisms
II. Inanimate (Concrete Categories)				
Plants	8	39	oak, pine	plants
Trees	37	80	maple, sequoia	trees → plants
Shrubs	36	120	rose, lavender	shrubs → bushes → plants
Fungi	39	40	mushroom	fungi

(continues on next page)

Category	Num.	Pairs	Samples	Hypernym Chains
Fruit	35	37	apple, orange	fruit
Vegetables	36	36	carrot, tomato	vegetables
Sweets	39	80	chocolate	sweets → desserts
Furniture	33	35	chair, bed	furniture
Clothing	20	80	shirt, pants	clothing → garments
Shoes	18	36	sneaker, boot	shoes → footwear
Tops	15	70	sweater, blouse	tops → garments
Accessories	18	42	scarf, hat	accessories
Jewellery	23	24	ring, necklace	jewellery
Headwear	13	14	helmet, cap	headwear
Toys	34	36	doll, puzzle	toys
Shops	30	93	bakery, florist	shops → stores → buildings
Buildings	30	40	house, cabin	buildings
Monuments	35	74	statue, pyramid	monuments → landmarks
Appliances	25	50	oven, toaster	appliances → devices
Electronics	26	58	phone, laptop	electronics → devices
Instruments	39	40	guitar, violin	instruments
Vehicles	33	35	car, truck, van	vehicles
Shapes	26	54	circle, triangle	shapes → forms
44	**1,198**	**2,576**		

Table 3.3. Overview of all 44 semantic categories included in the taxonomy. Each category lists the number of lexical examples and total generated hyponym–hypernym pairs, along with representative samples and hypernym chains.

As seen in our prompts, one deliberate design choice was to include only single-word samples (e.g. *beagle* but not *golden retriever*). This choice was motivated, first, by the fact that single words would be easier to pluralize consistently during the template-generation phase. Second, we believe that this property makes MoRe a more appealing resource for interpretability-focused experiments, that usually require precision at the lexical level. For example, recent mechanistic interpretability studies opt for single-word samples when investigating model internals (Wang et al., 2022). Multi-word expressions introduce additional complexity, as they are more likely to be split into multiple tokens or subtokens after tokenization. For this reason, since it remains unclear if longer spans might affect how information is represented and localized within a model, we decided to restrict MoRe to single-word lexical samples, to provide a controlled setting for future interpretability analyses.

It should be noted that before turning to prompt-based generation, we initially experimented with WordNet, a large-scale lexical database encoding hierarchical relations between nouns, verbs, and adjectives (Miller (1995). WordNet includes a large number of is-a relations, rendering it a popular choice among datasets that model semantic hierarchies, however, its use raised certain methodological challenges in the present experimental setting. Using WordNet made it, for example, difficult to follow our single-token constraint, considering that the database contains numerous multi-word entries whose pre-processing proved challenging. We thus frequently obtained truncated pairs (e.g. blue whale → blue, golden retriever → golden), which would require considerable post-editing to be filtered out. In addition, we observed that particularly inanimate categories were notably underrepresented, so we would likely need to combine additional sources to achieve the desired category sizes. As mentioned in previous work, integrating lexical relations from databases can also lead to unnatural sentences, due to the potential mismatch between a word's sense and the sentence context in which it is embedded (Glockner et al., 2018). For the aforementioned reasons, prompting GPT-2 helped us meet the dataset's criteria more easily, without the need for thorough post-processing. As shown in Table 3.3, the number of samples and consequently pairs varies slightly per category, something that we attribute to factors such as the exclusion of multiword instances as well as the fact that ChatGPT often produced invalid tokens that had to be removed.

3.3 Template Generation

Upon collecting our lexical pairs, we proceeded to the template design phase, where we prompted ChatGPT to produce a bank of sentence templates. For illustration, we provide below the exact prompt used for the category *mammals* under the operator *all*, followed by the resulting output:

Template Prompt for *mammals*

Generate 20 event-sounding sentence templates for the category *mammals* using the universal quantifier *all*. Each sentence must:

1. begin with "All {X} ..." where {X} is a plural noun placeholder;
2. describe an event, not a fact or general property, so avoid factual claims;
3. be syntactically simple (no clauses);
4. be short, clear, and compatible with any member of the category inserted in the placeholder;
5. follow a similar length across all 20 templates;
6. return the output as a Python list of 20 strings.

Sample Output

```
ALL = {
    "Mammals": [
        "All {X} moved across the field.",
        "All {X} rested near the trees.",
        "All {X} gathered by the river.",
        "All {X} searched for food.",
        "All {X} stayed close to the group.",
        "All {X} appeared near the village.",
        "All {X} were seen at dawn.",
        "All {X} returned after the sunset.",
        "All {X} stayed hidden in the dark.",
        [...]
    ]
}
```

As shown in the prompt, our templates were generated separately for each operator type. In practice, this means that the templates used for the "all" (R1–R2), "some" (R3–R4), "no" (R5–R6), and no-operator (R7–R8) rules share the same sentences and hypernym-hyponym pairs, differing only in the position of the latter (depending on whether they appear in the the premise vs. the hypothesis). Due to reasons of template economy, we did not design a separate set of sentences for the contradiction cases (R9-12), but we instead recombined the templates used for the other three operator types. For each of the 44 semantic categories in our taxonomy, we generated 30 category-compatible templates per operator, yielding a total of 5,280 unique templates[4].

We asked ChatGPT to generate sentences that are (ii) syntactically simple and (iii) of relatively uniform sentence length, as previous work shows that sentence length can become a spurious signal that models exploit (Gururangan et al., 2018). Additionally, we wanted our templates to describe events (e.g. "All mammals moved across the field") rather than factual

[4]calculated as 44 categories × 30 templates per category × 4 operators.

properties (e.g. “All mammals produce milk to feed their young”). Although we initially experimented with templates that express properties of the category members, we concluded that these sentences generalized poorly across different levels of the taxonomy. For example, templates such as “All dogs have fur” or “All dolphins have fins” do not scale to broader categories like mammals (not all mammals have fur) or animals (not all animals have fins). It should be noted that from a theoretical perspective, the task of NLI is independent of the pragmatic plausibility or factual truth of the sentences involved. However, since MoRe is a diagnostic dataset, we preferred to maintain control over factuality, considering that we cannot be certain about the extent to which a model’s decision can be affected when sentence meaning clashes with standard world knowledge that models acquire during pre-training.

3.4 Dataset Statistics

As seen in Table 3.4 MoRe contains a total of 927,360[5] premise-hypothesis pairs, corresponding to 77,280 instances per inference rule. The premises and hypotheses have an almost identical mean length of 6.3 tokens, a property that we tried to control for in our prompt in order to minimize the potential effect of stylistic variation in our experiments. The overall structure of MoRe is summarised in Table 3.5. In the next section, we present in detail how we organised MoRe into three experimental splits, each designed as a controlled test of generalization.

Property	**Value**
Mean premise/hypothesis length	6.3
Total number of semantic categories	44
Total number of unique hypernyms	62
Total number of hyponyms	1,198
Total hyponym–hypernym pairs	2,576
Total samples per rule	77,280
Label distribution (ent/neutr/contr)	309,120
Total dataset size	927,360

Table 3.4: Dataset Statistics.

Column	**Description**
`id`	Unique numerical identifier for each generated instance.
`gold_label`	entailment, neutral, contradiction
`rule`	Rule number (i.e. R1–R12)
`operator`	all, some, no, no-all, all-no, no-some, some-no
`monotonicity`	upward, downward, non-monotonic, or negation
`premise`	E.g. All birds gathered on the branch
`hypothesis`	E.g. All sparows gathered on the branch
`category`	Semantic class of the noun pair (e.g. *birds*)
`hypernym`	The higher-level lexical item (e.g. *birds*)
`hyponym`	The lower-level lexical item (e.g. *sparrows*)

Table 3.5: Columns included in MoRe.

at will allow us to conduct more targeted cross-rule experiments in subsequent stages.

[5]The final size of MoRe can be calculated by combining (i) the number of hyponym–hypernym pairs present in the taxonomy with (ii) the number of templates/category that were generated for each rule, yielding a total of: 2,576 pairs × 30 templates × 12 rules = 927,360 samples.

3.5 Experimental Splits

In the previous chapter, we explained that one important criterion for assessing robustness in language models concerns the ability to generalize systematically to novel combinations of familiar linguistic information, beyond what is explicitly seen in the training (Hupkes et al., 2020). Prior work has pointed out that a necessary prerequisite for assessing this type of generalization is "principled control over dataset syntax and semantics" (Hupkes et al., 2020), otherwise it becomes impossible to identify cases where models rely on unintended strategies to achieve high performance (Vegner et al., 2025, p. 5). We thus departed from the conventional practice of randomly splitting a dataset into training and test distributions and instead organized MoRe into three experimental splits of progressive difficulty. Each split corresponds to a systematic generalization setting where we carefully manipulate the lexical and syntactic patterns presented to amodel during training. Table 3.6 provides an overview of the design characteristics of each split, while Figure 3.5 provides examples from each experimental setting.

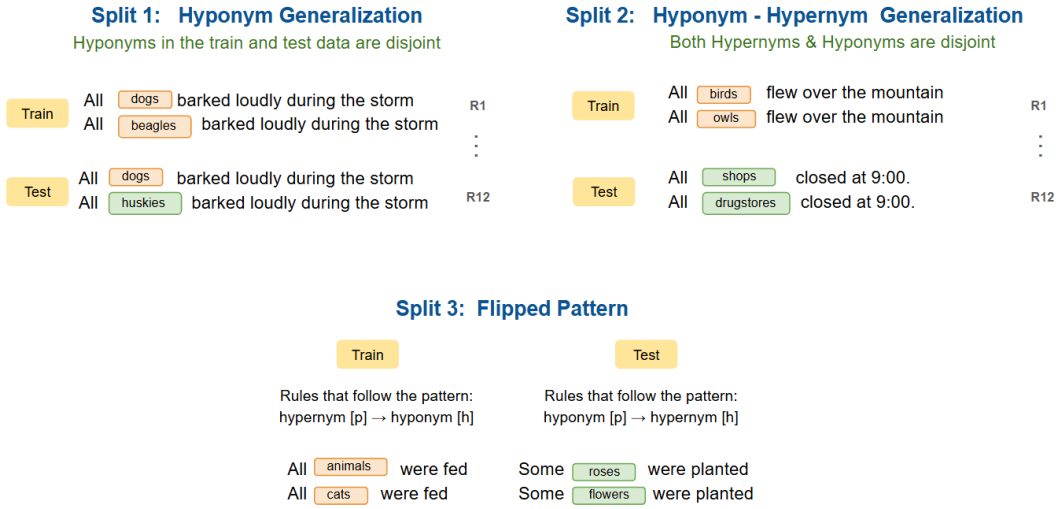


Figure 3.5: Experimental Splits

3.5.1 Split 1: Hyponym Generalization

In our Hyponym Generalization split, we expose a model to the full set of monotonic patterns contained in our dataset, but create a setting where the hyponyms (i.e. the subordinate terms) between the train and test data are kept disjoint. In practice, this means that the training and test data share all 12 monotonicity patterns (presented in Table 3.1), the same hypernyms and the same sentence templates. A clear illustration of this split is presented in Figure 3.5. Based on this design, this test can be used to assess two possible model behaviors. First, a fine-tuned model may capture the 12 monotonic patterns, encode information about hypernymy relations from training, and rely this information to generalize to unseen hyponyms at test time. On the other hand, this test contains a potential shortcut solution: since the premises between the training and test sets are essentially identical, a model may rely on this high lexical overlap as a shortcut for a label. For this reason, we accompany this split with a diagnostic variant that helps distinguish between these two possibilities. We discuss this diagnostic in Section 5.2.1.

3.5.2 Split 2: Hyponym-Hypernym Generalization

As illustrated in Figure 3.5, our Hypernym-Hyponym Generalization split replicates the aforementioned logic, but introduces a more demanding generalization setting where the lexical

Factor	Split 1	Split 2	Split 3
Monotonicity rules	✓	✓	✗
Hypernyms	✓	✗	✗
Hyponyms	✗	✗	✗
Sentence contexts	✓	✗	✗

Table 3.6: Overview of the dimensions held constant across the train and test distributions in MoRe splits.

overlap is removed completely. Specifically, we made sure that the hypernyms (i.e. the superordinate), the hyponyms (i.e. the subordinate) and the sentence templates used in the training set are now distinct from those encountered at test time[6]. For instance, this means that a model may be trained on an pattern containing a pair such as *birds–owls*, and be tested on a structurally familiar pattern containing an unseen pair (e.g. *shops–drugstores*). Given this design, we make two assumptions about a model's performance in this setting. First, since finetuning is generally effective at adapting models to dataset-specific regularities, we expect that an NLI model has the potential to capture the 12 monotonic patterns. Second, we hypothesize that high performance in this split can only be achieved by a system that encodes and relies on information about hypernym–hyponym relations. Because the test set introduces lexical pairs that never appear during training and there is no other apparent surface cue to rely on, a model that merely memorizes lexical pairs from the training set, or lacks information about semantic relations should perform poorly under these conditions.

3.5.3 Split 3: Flipped Pattern

One particularly promising application of MoRe is the study of shortcut learning. As a first step towards exploring this potential, we designed our third experimental split as a shortcut introducing setting that helps diagnose a system's behavior in the presence of an extreme shortcut association. To this end, we create a scenario where the pattern-label mappings between the train and the test data are reversed. In particular, the training distribution of this split includes monotonic patterns that teach a model the association "operator + hypernym in the premise + hyponym in the hypothesis = entailment / neutral"[7], yet testing is carried out on patterns that reverse this association: "operator + hyponym in the premise + hypernym in the hypothesis = entailment / neutral "[8]. For instance, as shown in Figure 3.2, a model might see an example such as *[P] All animals were fed [H] All cats were fed = entailment*. At test time, however, we "trick" the model by evaluating it on the reversed version of this pattern (*[P] All beagles slept [H] All dogs slept = neutral*), where flipping the direction of the lexical relation results in a new inference label.

In principle, to succeed in this test, a system needs to reverse the statistical patterns that it was optimized to learn. This, however, constitutes a form of logical generalization that cannot be derived directly from the surface statistical regularities of the training distribution. Overall, we hypothesize that a behavior as such is unlikely to emerge for three main reasons. First, prior work suggests that abstract reasoning, in the sense of deriving logical properties from examples (e.g., *All* $x \Rightarrow$ *All* y when $x \subset y$) and recombining them in novel ways (e.g., *All* $y \not\Rightarrow$ *All* x), is not an emergent ability of the current general systems (Talmor et al., 2020; Goodwin et al., 2020; Pavlick, 2022; Zhang et al., 2022; Chen and Gao, 2023). Second, fine-tuning is generally understood as the process of adapting a system to statistical patterns present in data, while

[6]To achieve a clean separation between the lexical pairs, we used the 44 semantic categories as hypernyms (presented in Table 3.3, under the column *Categories*) and divided them in two sets of 22 categories each.

[7]R1, R4, R5, R7.

[8]R2, R3, R6, R8.

Metric	Hyponym Gen.[1] Train	Hyponym Gen.[1] Test	Hypo-hyper Gen. Train	Hypo-hyper Gen. Test	Flipped Pattern Train	Flipped Pattern Test
Unique hypo/s	599	599	642	556	642	556
Unique hyper/s	62	62	22	22	22	22
Unique pairs	1,274	1,302	642	556	642	556
Size	458,640	468,720	231,120	200,160	96,300	83,400

Table 3.7: Comparison of dataset statistics acrossour three splits for both train and test partitions. The labels (entailment, neutral, contradiction) remain balanced across Splits 1 and 2. Since the Flipped Pattern split 3 divides the training and test data per inference pattern, the label distribution is inevitably unbalanced (i.e. neutral: 50,040 , entailment: 16,680, contradiction: 16,680).

the impressive performance of finetuned LLMs is attributed exactly to their ability to capture and reproduce such patterns (Bender et al., 2021). Third, the training distribution of this split provides no opportunity a model to overcome a shortcut. Taken together, these hypotheses lead us to expect that a model will continue to apply the pattern-label mapping learnt during training, rather than invert it. Consequently, it will systematically predict the wrong label and thus fail to generalize.

[1]In our experiments, we treat the test distribution of the Hyponym Generalization split as the main test split of MoRe.

Chapter 4

Experimental Setting and Methodology

4.1 Motivation

When an LLM is evaluated on a reasoning task such as Natural Language Inference (NLI), its performance is influenced by two primary factors. First, its pre-training on large-scale, heterogeneous data, a stage where models implicitly encode statistical regularities about language in the form of latent, high-dimentional representations. Yet, since LLMs are opaque, and often the exact composition of their training corpora is obscured, it is not possible to precisely determine what task-specific knowledge a system possesses prior to task-adaptation. The second influence stems from fine-tuning. At this phase, a model is exposed to labeled data and adapts to the specific statistical regularities that underlie these data. One view is that the impressive performance of LLMs is a sophisticated, yet shallow, form of pattern matching Bender et al. (2021). Another perspective is that scaling laws alongside with more nuanced training methods, can unlock emergent abilities that go beyond mere pattern recognition Brown et al. (2020).

Following Zhang et al. (2022), an important distinction that is often neglected in the computational study of such phenomena, is that "learning to reason" and "learning to solve a reasoning benchmark" are two entirely different skills. With respect to the former aspect, the majority of existing evidence suggests that this form of logical flexibility traced in human reasoning remains beyond the reach of current language models (Talmor et al., 2020; Zhang et al., 2022; Pavlick, 2022; Mondorf and Plank, 2024), at least when training relies on standard, fine-tuning techniques (Chen and Gao, 2023). The capacity for abstract reasoning, as understood in classical symbolic artificial intelligence (i.e. formal logic) is not present in the design objectives of large language models, and has not convincingly emerged from large-scale pre-training (Liu et al., 2022; Creswell et al., 2022; Matthew Lam et al., 2024; Farjami et al., 2026). Nevertheless, the high-benchmark performance that LLMs achieve on logical benchmarks, when trained on sufficiently representative data (Yanaka et al., 2019a,c; Geiger et al., 2020), suggests two important things: first, that general systems can be steered to learn task-relevant properties, and second, that they likely possess useful linguistic representations that make such adaptation possible.

4.2 Models

For the scope of this thesis, we use small and computationally efficient baselines, as our primary goal is to investigate the diagnostic properties of the dataset. To this end, we selected two transformer-based models, `bert-base-uncased` (Devlin et al., 2019) and GPT-2 (Radford et al., 2019). In Sections 4.2.1 and 4.2.2, we describe how we adapted the two models for NLI classification. All experiments conducted for this thesis were conducted using Colab Pro, on an

Model	Architecture	Layers	Hidden Size	Heads	Vocab Size	Parameters
BERT-base	Encoder-only	12	768	12	∼30.5K	110M
GPT-2	Decoder-only	12	768	12	∼50K	117M

Table 4.1: Architectural characteristics of the two models

Model	Optimizer	LR	Epochs	Batch size	Seed	Metric
BERT-base	AdamW	2e-5	1	32	42	Accuracy
GPT-2	AdamW	2e-5	1	64	42	Accuracy

Table 4.2: The fine-tuning hyperparameters employed to train BERT-base and GPT-2 in our experiments.

L4 GPU, with a total estimated training time of approximately 7 hours.[1]

4.2.1 Training BERT

BERT-base (Devlin et al., 2019) is an encoder-only Transformer with 12 layers, 12 self-attention heads per layer, and a hidden dimensionality of 768, resulting in approximately 110 million parameters. As discussed in earlier chapters, BERT has demonstrated strong performance on NLI (Williams et al., 2018; Talman and Chatzikyriakidis, 2019) and Monotonicity (Yanaka et al., 2019a, 2020; Geiger et al., 2020; Rozanova and Pavlick, 2021; Wijnholds, 2023), therefore its use helps us to situate our findings within the broader literature.

To train BERT for NLI classification, we followed the standard finetuning procedure proposed by Devlin et al. (2019). In particular, we formatted each premise–hypothesis pair as a single input sequence using the special `[CLS]` and `[SEP]` tokens, as for instance in: "`[CLS]` All dogs are mammals `[SEP]` All beagles are mammals [SEP]". After passing through BERT's 12 Transformer layers, the final contextualized representation corresponding to the `[CLS]` token of each pair is then passed to a linear classification layer, where the model predicts the most probable of the three available classes (i.e. entailment, neutral, or contradiction). As illustrated in Table 4.2, we used standard hyperparameter settings proposed in the original paper (Devlin et al., 2019), across all experiments, including an effective batch size of 64, a context length of 128, and a fixed random seed for reproducibility. Additionally, we opted for a conservative learning rate, considering that larger learning rates have been associated with overfitting behaviours as well as phenomena such as that of catastrophic forgetting (Sun et al., 2020). Since finetuning is a computationally expensive technique, we trained BERT for a single epoch. Given that our training splits are large, ranging from approximately 100,000 to over 400,000 instances, we expect that one pass over the training data provides a sufficient learning signal over the inference patterns of our dataset.

4.2.2 Training GPT-2

The second model we employed is GPT-2 small (Radford et al., 2019) a decoder-only generative transformer comprised of 12 layers, 12 self-attention heads, and 117 million parameters. Despite being the smallest variant of the GPT family, GPT-2 remains a popular architectural choice in interpretability research, where a common methodological strategy is to begin with smaller models before scaling to larger architectures (Olah et al., 2020; Elhage et al., 2022). For instance, GPT-2 continues to serve as a standard model in recent studies centered around tracing circuits, disentangling feature representations and evaluating neuron-level behavior (Wang et al., 2022; Liu and Variengien, 2023; Zhang et al., 2025; Heimersheim, 2024; Eshuijs et al., 2025), as it offers a good balance between efficiency and feasibility. On the other hand, monotonicity reasoning in

[1] Our code is available at this repository.

NLI has predominantly been investigated using encoder-based architectures, especially BERT and RoBERTa models (Yanaka et al., 2019b; Richardson et al., 2019; Rozanova and Pavlick, 2021). NLI has been characterized as a demanding task for autoregressive models (Liu et al., 2023b), and for this reason, existing studies suggest that fine-tuning them on NLI constitutes an interesting but underexplored research direction (Brown et al., 2020).

We trained GPT-2 on our dataset in two different ways. First, we fine-tuned the model employing the methodology and code made available by Eshuijs et al. (2025), using almost identical hyperparameters to those of BERT (presented in Table 4.2). GPT-2 is a unidirectional model, meaning that it predicts the next token attending solely on the left context. To train GPT-2 as an NLI classifier, we insert each premise–hypothesis into a prompt template, and the model is optimized on the task by employing its standard next-token prediction objective. To obtain GPT's prediction, instead of asking it to generate a label, we extract the logits corresponding to the three label tokens (A, B, and C) and select the label that received the highest confidence score at test time[2]

```
Perform NLI on the following sentences:

Sentence 1: {premise}
Sentence 2: {hypothesis}

Label Options: A: entailment   B: neutral   C: contradiction

Label:
```

GPT-2 was also trained under a zero- and a few-shot prompting conditions. In-Context-Learning (ICL) is a promising ability of autoregressive models like GPT. Much recent evidence has shown that well-designed prompts, often containing labeled examples as in Few-Shot-Learning (FSL), can help a model solve a task, overcoming the need for extensive training and further parameter updates (Brown et al., 2020). Importantly, since models are tested on data that has not been explicitly seen during training, this setting provides a clearer picture of whether model representations are useful for the intended task (Talmor et al., 2020). The prompt used in our experiments is presented in Table 4.3. For the zero-shot setting, GPT-2 was given only the instructions, while for the few-shot setting, we provided 12 labeled premise-hypothesis pairs, representative of the monotonic patterns presented at test time.

Zero/Few-Shot Prompt

Determine the logical relationship between the following premise and hypothesis for the task of NLI. Choose one label: `entailment`, `neutral`, or `contradiction`.

Examples:

Premise: `All insects crawled under the leaves.`
Hypothesis: `All ants crawled under the leaves.`
Answer: `entailment`

Premise: `All parrots rested on the branches.`
Hypothesis: `All birds rested on the branches.`
Answer: `neutral`

`[....]`

Now classify the following pair:
Premise: `<premise>`
Hypothesis: `<hypothesis>`
Answer:

Table 4.3: The Zero- and Few-shot Prompt Template used to train GPT-2

[2]Eshuijs et al. (2025)'s code is available at this link.

4.3 Experimental Setting

As discussed in Chapter 1, monotonicity constitutes a cognitively demanding form of reasoning that does not emerge robustly in current NLI systems as a systematic capability (Yanaka et al., 2019a; Chen and Gao, 2023). However, both in the case of NLI and in many other NLP tasks, it is well established that Supervised Finetuning (SFT), constitutes an effective method for adapting general language systems to specific downstream tasks. What is often neglected, and what drives the methodology of this thesis, is that transformer-based classifiers remain quite vulnerable to distribution shifts. In this sense, their potential to generalize depends decisively on whether the training data provides them with sufficient learning signals to support the desired behaviour (Geiger et al., 2019).

Against this background, we designed our experiments in line with two broad hypotheses. *Hypothesis 1:* if the data provides all the necessary statistical cues for generalization, a model will likely capture and replicate them successfully on statistically similar instances. Existing evidence shows that model representations encode useful linguistic information about syntax (Jawahar et al., 2019; Tenney et al., 2019; Linzen and Baroni, 2021; Mahowald et al., 2024) as well as lexical relations such as that of hypernymy and hyponymy (Ettinger, 2020; Vulić et al., 2020; Rozanova et al., 2022; Regneri and et al., 2024; Diera and Scherp, 2026), which constitute the fundamental elements of monotonic logic. We therefore hypothesize that the two models we employ in our experiments contain the necessary information to pick up the reasoning patterns of MoRe and generalize them to structurally identical patterns that contain unseen elements, as occurs in our Hyponym Generalization (Section 3.5.1) and Hyponym-Hypernym Generalization splits (Section 3.5.2). *Hypothesis 2:* when an evaluation requires applying an abstract reasoning principle that cannot be derived directly from the surface statistical regularities observed during training, a system is expected to fail. This hypothesis is the theoretical motivation behind the design of our Flipped Pattern Split (Section 3.5.3), where the solution to the task requires reversing the pattern learnt during training rather than simply reproducing it.

4.3.1 Research Questions

At a first level, the design of MoRe allows us to address a primary, data-oriented research question: *Can GPT-2 and BERT capture and generalize over the monotonic logic encoded in MoRe?* To address this question, we conduct a series of empirical experiments that each assesses our models' generalization performance under a series of distribution shifts. A summary of our experiments is detailed in Table 4.4.

(i) *Can monotonic reasoning emerge from GPT-2's pretraining through zero and few-shot prompting?*

In our first experiment, we examine whether GPT-2 can handle monotonic reasoning in a zero- and few-shot setting, using the prompt presented in Table 4.3.[3] As evaluation data, we use the test set of SNLI (Bowman et al., 2015) as well as the test distribution of our Hyponym Generalization split (presented in Table 3.7). It should be noted that for the experiments conducted in this thesis, we treat the test distribution of the Hyponym Generalization split as the main test split of MoRe. Overall, we treat this experiment as a "vanilla" evaluation setting. Since we do not train GPT-2 on either dataset, its performance under these conditions can show more clearly whether the model already contains sufficient information to handle general and monotonic NLI. Despite the promising potential of In-Context-Learning (ICL) (Brown et al., 2020), empirical evidence suggests that NLI remains a challenging task for prompting-based approaches. For instance, Brown et al. (2020) show that few-shot GPT-3 underperforms fine-tuned BERT-large on NLI tasks included in the SuperGLUE benchmark. Additionally, few-shot GPTs perform at random chance on popular NLI datasets like MNLI and ANLI (Talman and

[3]For the zero-shot evaluation of SNLI and Split 1 we only use the instructions presented in the prompt. For the few-shot evaluation of Split 1, we also include 12 monotonic examples.

Model(s)	Method	Train	Test	Aim
GPT-2	zero/few-shot	–	Split 1, SNLI	Can GPT2 handle general and monotonic NLI with no prior training?
GPT-2	fine-tuning	SNLI	Split 1, SNLI	Does training on general NLI transfer to monotonic NLI?
GPT-2, BERT	fine-tuning	Split 1	Split 1	Can the models generalize to unseen hyponyms?
GPT-2, BERT	fine-tuning	Split 2	Split 2	Can the models generalize to unseen hyponym/hypernyms?
GPT-2, BERT	fine-tuning	Split 3	Split 3	Can the models reverse a reasoning pattern?

Table 4.4: Summary of the experiments conducted in this thesis. By Split 1, we refer to the Hyponym Generalization split; by Split 2, the Hypernym–Hyponym Generalization split; and by Split 3, the Flipped Pattern split.

Chatzikyriakidis, 2019; Du et al., 2023b). Based on these findings, we expect that GPT-2 will likely struggle to perform well in our ICL setup.

(ii) *Does fine-tuning on general NLI (SNLI) enable out-of-distribution generalization monotonic inferences?*

As discussed in Section 2.2.2, much prior work on monotonicity addresses the question of whether NLI models trained on broader natural language inference benchmarks can generalize to monotonic inferences (Yanaka et al., 2019b,a; Richardson et al., 2019; Rozanova et al., 2022). Following this evaluation paradigm, we finetune GPT-2 on SNLI (Stanford Natural Language Inference) (Bowman et al., 2015) and test it on our Hyponym Generalization split. Although SNLI and MoRe target the same task and share the same label space (entailment, neutral, contradiction), they differ in the types of inferences they contain. SNLI is a large-scale benchmark consisted of human-written premise-hypothesis pairs derived from image captions. The dataset contains heterogeneous inference phenomena, the majority of which rely on common sense reasoning (e.g. [P] A dog jumps to his owner. [H] A dog is with his owner. [ent.]). Previous work suggests that the transfer from general NLI to monotonic inferences is generally poor (Geiger et al., 2020; Rozanova et al., 2022; Chen and Gao, 2023; Wijnholds, 2023). At the same time, there is evidence that models trained on SNLI acquire some information on lexical relations, however, this knowledge alone appears insufficient for monotonicity reasoning involving logical operators (Geiger et al., 2020). With this in mind, we treat this setup as a standard evaluation that will help us assess whether the limitations reported in previous work are likewise observed on MoRe.

(iii) *Can models generalize to unseen hypernym–hyponym substitutions and templates?*

In our third experiment, we finetune GPT-2 and BERT-base on the Hyponym Generalization and Hypernym-Hyponym Generalization splits. As explained in detail in Sections 3.5.1 and 3.5.2, both splits expose the models to all 12 monotonicity patterns present in the dataset. The first split evaluates whether a model can generalize these patterns to examples containing unseen

hyponyms. The second split presents a more challenging setting in which both hypernyms, hyponyms and sentence templates are unseen at test time. At their core, both splits address the same question: *can models apply familiar monotonicity patterns to novel lexical instantiations of those patterns?*. For this reason, we hypothesize that our fine-tuned models can perform well in these two experiments, if they encode information about semantic relations, which is a necessary prerequisite for this type of generalization.

(iv) *Can we teach a model a shortcut?*

In this experiment, we fine-tune GPT-2 and BERT-base on our Flipped Pattern split. As explained in Section 3.5.3, this test creates an extreme shortcut condition where a model is trained and evaluated on reversed pattern-label associations. We designed this extreme, yet useful diagnostic scenario to isolate the effects of shortcut learning as clearly as possible. We hypothesize that when a fine-tuned system is trained on a fully biased shortcut signal, it is expected to capture and reproduce it independently of whether this association is the solution intended by the test data. Since the shortcut becomes invalid at test time, we expect that both GPT-2 and BERT will fail to generalize in this setting.

Chapter 5

Results

To establish a baseline for GPT2's capacity to solve NLI without prior fine-tuning, we evaluated the model under *zero-shot* and *few-shot* prompting conditions (our prompts are presented in Chapter 4, Figure 4.3). As seen in Table 5.1, the two prompting settings led to a near-chance accuracy both for SNLI and MoRe. These initial findings suggest that GPT-2 does not possess enough information to perform general and Monotonic NLI without further training. This result confirms our expectation that Monotonicity and the task of Natural Language Inference in general, are demanding logical phenomena that do not emerge spontaneously from GPT-2's pretraining (Talman and Chatzikyriakidis, 2019; Chen and Gao, 2022; Du et al., 2023a; Chen and Gao, 2023). Moreover, the model did not learn how to solve monotonic inferences effectively from the labeled premise-hypothesis pairs provided in the few-shot prompt, suggesting that supervised fine-tuning is necessary to teach GPT-2 the reasoning patterns contained in our dataset. It should be noted that even when we conducted preliminary experiments with more detailed, or chain-of-thought-style prompts, we saw no improvement in GPT-2's performance. We therefore hypothesize that the low accuracy observed in both settings is more likely attributable to the model itself, rather than than to limitations of our prompt formulation. Yet our experiments provide a useful baseline, and leave the exploration of more nuanced prompting strategies as an interesting direction for future work.

Model	**Test Data**	**Pairs**	**Accuracy**	**Precision**	**Recall**	**F1**
GPT-2 zero	SNLI	10k	33.1	16.7	25.6	18.1
GPT-2 zero	Hyponym Gen Split	468k	31.3	20.9	31.3	24.4
GPT-2 few	Hyponym Gen.Split	468k	34.9	24.1	34.9	24.1

Table 5.1: Zero- and few-shot performance of GPT-2 on SNLI and the MoRe Hyponym Generalization split.

	Entailment			**Neutral**			**Contradiction**		
Experiment	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
SNLI Zero-shot	0.0	0.0	0.0	31.5	32.4	32.0	31.2	61.6	41.4
MoRe Zero-shot	0.0	0.0	0.0	31.5	32.4	32.0	31.0	61.0	41.0
MoRe Few-shot	0.0	0.0	0.0	38.1	16.3	22.9	34.4	88.5	49.6

Table 5.2: Per-label precision, recall, and F1-score of GPT-2's predictions on SNLI and MoRe. The scores are aggregated across rules sharing the same NLI label.

To gain a better understanding of GPT-2's misclassifications, we conducted a per-label analysis on both SNLI and MoRe. It should be noted that for all the per-rule analyses, we report only accuracy, as each monotonicity rule is associated with a single gold label. The findings reported

Label	Rule	Zero-GPT	Few-GPT
Ent.	R1 (All animals ate → All horses ate)	0.0	0.0
	R3 (Some poodles ran → Some dogs ran)	0.0	0.0
	R5 (No shops were open → No bakeries were open)	0.0	0.0
	R7 (The shoes were wet→ The sneakers were wet)	0.0	0.0
Neut.	R2 (All horses ate → All animals ate)	23.9	$56.5^{\uparrow}$
	R4 (Some dogs ran → Some poodles ran)	55.5	$3.8^{\downarrow}$
	R6 (No bakeries were open → No shops were open)	36.8	$2.6^{\downarrow}$
	R8 (The sneakers were wet→ The shoes were wet)	13.4	$2.5^{\downarrow}$
Contr.	R9 (All birds flew → No sparrows flew)	69.7	$77.7^{\uparrow}$
	R10 (No sparrows flew → All birds flew)	77.5	$82.4^{\uparrow}$
	R11 (No flowers bloomed → Some roses bloomed)	47.6	$99.8^{\uparrow}$
	R12 (Some roses bloomed → No flowers bloomed)	51.4	$94.2^{\uparrow}$

Table 5.3: Per-rule accuracy of GPT-2 on MoRe's Hyponym Generalization test data split in our zero-shot and few-shot prompting experiments. A complete overview of the 12 monotonicity patterns can be found in Table 3.1.

in Table 5.2 demonstrate that GPT-2's performance is unsymmetrical across the three NLI labels. A notable observation is that GPT-2 tended to over-predict contradiction across the entire dataset, resulting in a large number of entailment and neutral instances being misclassified as contradiction. This tendency is reflected in GPT-2's relatively high recall for contradiction across all experiments (61%–88.5%), indicating that the model identified a considerable proportion of true contradiction inferences. At the same time, the low precision of this label (31.2%, 31.0%, and 34.4%) shows that many instances predicted by GPT-2 as contradiction actually belonged to the entailment or neutral classes. The confusion matrix, presented in Appendix A confirms this tendency. Upon conducting a more detailed rule-level analysis on GPT-2's predictions on MoRe (presented in Table 5.3) we did confirm that GPT-2 scores higher on those monotonic patterns that result in a contradiction label (i.e. Rules 9–12), and this performance improved further through the few-shot prompt. While we cannot precisely determine the exact source of this bias or why GPT-2, we hypothesize that the model might rely on shortcuts associated with this class. For instance, the presence of the operator "No" in these rules (e.g. *R9:* [P] All birds flew [H] **No** birds flew, *R10:* [P] **No** birds flew [H] All birds flew) might have acted as a shortcut signal for contradiction. Indeed, prior studies demonstrate that negation words (e.g. *no. not, never*) often act as contradiction shortcuts by NLI models (Naik et al., 2018; Gururangan et al., 2018; McCoy et al., 2019). However, the fact that GPT-2 tends to default to contradiction globally, may also reflect an internal statistical bias inherited from its pretraining. If this is the case, we expect that a similar trend will likely emerge in subsequent experiments as well.

As seen in Table 5.2, GPT-2 failed to predict entailment in both datasets, even when it was provided with labeled entailment demonstrations (e.g. [P] All dogs slept, [H] All beagles slept = *ent.*) in the few-shot setting. On the other hand, while GPT-2 classified some neutral inferences correctly, its performance on this inference relation remained close to chance level.

5.1 Out-of-Distribution Evaluation from General to Monotonic NLI

Our previous experiments suggest that GPT-2 encodes limited knowledge of both general and monotonic NLI, as evidenced by its poor zero- and few- shot performance on SNLI and MoRe. At the same time, in-context examples did not help the model learn to solve the task of NLI, a finding that we find rather unsurprising. As discussed in the previous chapter, linguistic

reasoning is not viewed as an emerging ability of GPT-2. Consequently, prior studies argue that the acquisition of such complex logical reasoning skills is unlikely to emerge from instructions or a small number of labeled demonstrations, particularly in relatively small foundation models such as GPT-2 (Chen and Gao, 2023).

Model	Test Data	Accuracy	Precision	Recall	F1
GPT-2 - SNLI	SNLI test	88.3	88.0	88.3	88.1
	Hyponym Gen. test	34.6	32.5	34.5	32.8

Table 5.4: Performance of fine-tuned GPT2 on SNLI, on the test distribution of SNLI and the standard test split of MoRe (Hyponym Generalization Split.)

Against this background, we proceed with fine-tuning GPT-2, under the assumption that explicit training on NLI data will help the model adapt for NLI classification more effectively than the in-context-learning setup. In the present experiment, we fine-tune GPT-2 on the train distribution of SNLI, a general-purpose NLI dataset. Evaluation is conducted on on (i) an in-distribution setting (ID), namely the SNLI test partition, and (ii) an out-of-distribution setting (OOD), namely the test split of MoRe [1]. Our goal in this evaluation is two-fold. First, we examine whether fine-tuning can help GPT-2 capture useful linguistic information about inference, which we found to be weakly encoded in its pre-trained representations. This is primarily assessed through the in-domain experiment. Specifically, since the SNLI test partition is drawn from the same distribution as the training data, this setting can show more clearly if GPT-2 acquired general NLI skills during training. Second, we investigate whether training on a general inference benchmark such as SNLI can improve GPT-2's performance on monotonic reasoning, which we consider a more demanding evaluation of the model's reasoning and generalization abilities.

The results presented in Table 5.4 show that the SNLI-finetuned GPT-2 achieved a high in-distribution accuracy on the SNLI test split (88.3%). As expected, this result indicates that fine-tuning helped GPT-2 capture useful information required to solve SNLI. Yet, despite its strong in-domain performance, GPT-2 generalized poorly on the test split of MoRe, reporting an accuracy of only 34.6%. This finding confirms our expectation, as previous studies have shown that models trained on general NLI benchmarks fail to generalize to monotonic NLI (Geiger et al., 2020; Rozanova et al., 2021, 2022; Chen and Gao, 2023; Wijnholds, 2023). The most possible explanation behind this low transfer is that SNLI might not contain sufficient learning signals about monotonicity to allow for a more robust generalization to our dataset. For instance, Geiger et al. (2020) show that models trained on SNLI acquire some information on lexical relations, which is a core element of monotonic logic, yet this knowledge alone appears to be insufficient for handling monotonicity reasoning.

Table 5.5 provides a more detailed analysis of GPT-2's performance on MoRe, organised by label and monotonicity rule. Overall, the results reveal considerable variability in GPT-2's predictions. First, the SNLI-finetuned GPT-2 performed again comparatively better on contradiction patterns (i.e. R9-12). As for the remaining two classes, SNLI did not result in robust predictions for entailment and neutral monotonic patterns. According to Table 5.5, GPT seems to have picked up subtle knowledge about entailment from SNLI, given that the zero-shot baseline failed to predict this label completely. On the other hand, SNLI harmed performance on neutral inferences, as the accuracy of most neutral rules (R2-8) is close to zero. An interesting exception concerns Rules 3 and 4, which were the only non-contradiction patterns for which GPT-2 attained a moderately high accuracy (∼67%). To illustrate, Rule 3 corresponds to inferences of the form [P] Some + hyponym [H] Some + hypernym = entailment (e.g. *[P] Some poodles ran [H] Some dogs ran*), whereas Rule 4 instantiates cases that follow the pattern [P] Some + hypernym, [H] Some + hyponym = neutral (e.g. [P] *Some dogs ran [H] Some poodles*

[1]For the purposes of this thesis, we treat the test distribution of the Hyponym Generalization split (introduced in Section 3.5.1) as the main test split of MoRe.

Label	Rule	GPT-2 - SNLI Acc.
Entailment	R1 (All animals ate → All horses ate)	3.8
	R3 (Some poodles ran → Some dogs ran)	67.7
	R5 (No shops were open → No bakeries were open)	6.9
	R7 (The shoes were wet→ The sneakers were wet)	6.5
Neutral	R2 (All horses ate → All animals ate)	3.3
	R4 (Some dogs ran → Some poodles ran)	66.3
	R6 (No bakeries were open → No shops were open)	1.8
	R8 (The sneakers were wet→ The shoes were wet)	3.7
Contradiction	R9 (All birds flew → No sparrows flew)	44.4
	R10 (No sparrows flew → All birds flew)	80.9
	R11 (No flowers bloomed → Some roses bloomed)	33.3
	R12 (Some roses bloomed → No flowers bloomed)	96.3

Table 5.5: Per-rule Accuracy of the SNLI Fine-tuned GPT-2 on MoRe'es Hyponym Generalization Test Data Split

ran). With respect to Rule 3, the fact that GPT-2's performance increased from 0% in the zero-shot setting (as reported earlier in Table 5.3) to 68% after fine-tuning suggests that SNLI contains information that is useful for solving this particular inference pattern. Notably, given that Rule 3 is the only upward-entailing[2] pattern of MoRe, this isolated improvement appears to reflect previously documented evidence surrounding the distributional properties of SNLI. In particular, previous studies have shown that general NLI datasets such as SNLI contain a large number of upward-entailing premise-hypothesis pairs, leading models to perform comparatively better on this monotonic direction (Yanaka et al., 2019a; Wang et al., 2019b; Rozanova et al., 2021; Feng et al., 2022; Chen and Gao, 2022, 2023; Rozanova et al., 2024).

5.2 Evaluating GPT-2 and Bert-base on MoRe

So far, neither in-context learning nor SNLI-finetuning has resulted in robust performance on MoRe. In the following experiments, we therefore train GPT-2 and BERT-base directly on MoRe's experimental splits, to assess whether fine-tuning can help the two models capture linguistic information necessary to handle monotonic inferences. An overview of MoRe's splits can be found in Table 5.6

5.2.1 Evaluation of the Hyponym-Generalization Split

As seen in Table 5.7, both models achieved a near-ceiling accuracy on the Hyponym Generalization Split of MoRe. As illustrated in Tables 5.6 and 5.8, our Hyponym Generalization test operationalizes a very controlled condition where models are expected to generalize to structurally identical premise-hypothesis pairs, that nevertheless contain hyponyms that were not observed during training (e.g. *horses* in test vs. *sheep* in train). Given the design of this experiment,

[2]As explained in Chapter 1, a monotonic direction is characterized as upward when a subordinate term (e.g. poodle) followed by a subordinate term (e.g. dog) preserves the entailing relationship between the premise and the hypothesis.

[2]Since the Flipped Pattern split divides the training and test data per inference pattern rather than label, training contains only rules following the hypernym$_P$ → hyponym$_H$ pattern (R1, R4, R5, R7), while testing contains only rules following the hyponym$_P$ → hypernym$_H$ pattern (R2, R3, R6, R8). The label distribution is therefore necessarily imbalanced: the training set contains three entailment rules (R1,5,7) and one neutral rule (R4), whereas the test set contains one entailment rule (R3) and three neutral rules (R2,6,8), yielding a 3:1 and 1:3 entailment-neutral ratio respectively.

Experimental Split	Train	Test	Label Distr.
Hyponym Generalization			
All 12 patterns, sentence templates and hypernyms present in train and test, but hyponyms are disjoint.	458k	468k	Balanced (1:1:1)
Train: All *dogs* barked → All *beagles* barked (ent.) Test: All *dogs* barked → All terriers barked (ent.)			
Hypo-Hypernym Generalization			
All 12 patterns present in train and test but templates, hyponyms & hypernyms are disjoint.	231k	200k	Balanced (1:1:1)
Train: All birds flew → All owls flew (ent.) Test: All shops closed → All drugstores closed (neutr.)			
Flipped Pattern			
Test examples reverse the pattern observed during training. Both pairs and templates remain disjoint.	96k	83k	Imbalanced[2]
Train: $\text{hypernym}_P \rightarrow \text{hyponym}_H$ e.g. All animals$_{\text{hyper}}$ were fed → All cats$_{\text{hypo}}$ were fed (ent.)			
Test: $\text{hyponym}_P \rightarrow \text{hypernym}_H$ e.g. All huskies$_{\text{hypo}}$ slept → All dogs$_{\text{hyper}}$ slept (neutr.)			

Table 5.6: Overview of the three Experimental Splits of MoRe.

Model	Hyponym Gen.	Hypo-Hypernym Gen.	Flipped Pattern
GPT-2	99.9	87.4	0.0
BERT-base	99.9	96.6	0.0

Table 5.7: Aggregated Accuracies for GPT-2 and BERT-base on MoRe's Experimental Splits. The Evaluation of each Split will be discussed separately in the following Sections. The Full Classification Report for each Split can be found in Table A.2 in Appendix A.

we consider two possible explanations behind the near-perfect predictions of the two models. First, the fact that GPT-2 and BERT generalized successfully to unseen hyponyms could be interpreted as positive evidence that the models picked up the monotonic patterns of this split and learnt to encode information about hypernym-hyponym relations during fine-tuning. This particular split, as well as MoRe in general, provides a strong learning signal regarding how lexical substitutions interact with the rules of monotonic reasoning. Consequently, it is likely that after their extensive exposure to these patterns, the models identified hypernym–hyponym relations as a decisive feature for solving the task. According to prior evidence, language models encode information about the lexical relation of hypernymy already in their pretrained representations (Ettinger, 2020; Vulić et al., 2020; Rozanova et al., 2022; Regneri and et al., 2024; Diera and Scherp, 2026). It is therefore plausible that the training signal provided by MoRe helped the models exploit hypernymy-related information to solve the unseen hyponym test. On the other hand, the near-perfect accuracy could also suggest that the two models exploited the high lexical overlap that we intentionally maintained between the train and test data of this split as a form of shortcut. As shown in Table 5.8, by design, the premises and hypotheses appearing in the training and test set are essentially identical (e.g. Some animals run - Some

Hyponym Generalization Split	
Train	**Test**
R1 All animals ate → All horses ate [E]	**R1** All animals ate → All sheep ate [E]
[P] all + hypernym [H] all + hyponym = Ent.	[P] all + hypernym [H] all + hyponym = Ent.
[...]	[...]
R4 Some animals ran → Some poodles ran [N]	**R4** Some animals ran → Some huskies ran [N]
[P] some + hypernym [H] some + hyponym = Neutr.	[P] some + hypernym [H] some + hyponym = Neutr.
[...]	[...]
R12 Some roses bloomed → No flowers bloomed [C]	**R12** Some lilies bloomed → No flowers bloomed [C]
[P] some + hyponym [H] no + hypernym = Contr.	[P] some + hyponym [H] no + hypernym = Contr.

Table 5.8: Illustration of the Hyponym Generalization Split.

poodles run [ent.] vs. Some animals run - Some huskies run [ent.]), except for the hyponym present at test time. This means that in principle, a model that, for instance, merely memorizes a premise-label correlation from training could generalize almost perfectly without necessarily considering the hypernymy–hyponymy relations that the task is intended to evaluate. Considering that NLI models are highly prone to employing unintended strategies when making a prediction, we cannot exclude the possibility that our models leveraged the shared hypernyms of the train and test data as shortcuts. For example, McCoy et al. (2019) show that NLI models tend to overpredict entailment when the premise and hypothesis share a large number of overlapping words. An even more surprising case is the hypothesis-only bias (Poliak et al., 2018; Glockner et al., 2018; McKenna et al., 2023), as models have been found to achieve high classification accuracy without even seeing the premise. Therefore, this result alone cannot determine with certainty whether the models solved the task by capturing and applying the monotonicity patterns or by exploiting the hypernym heuristic present in the dataset.

Hyponym Generalization Split Diagnostic	
Train	**Test**
R1 All animals ate → All horses ate [E]	**R1** All organisms ate → All sheep ate [E]
[P] all + hypernym [H] all + hyponym = Ent.	[P] all + hypernym [H] all + hyponym = Ent.
[...]	[...]
R12 Some chairs broke → No furniture broke [C]	**R12** Some tables broke → No things broke [C]
[P] some + hyponym [H] no + hypernym = Contr.	[P] some + hyponym [H] no + hypernym = Contr.

Table 5.9: Illustration of the Diagnostic Version of the Hyponym Generalization Split. In this test, the train and test data contain disjoint hypernym-hyponym pairs, yet the same sentence templates.

For this reason, we designed an additional diagnostic experiment where the hypernym shortcut is removed. To achieve that, we replaced all hypernyms in the test set of this split with unseen hypernyms, while keeping the sentence templates and the underlying monotonicity rules unchanged. As illustrated in Table 5.9, the models are presented with the same training data (e.g. *All animals ate - All horses ate* [ent.]), yet this time, we insert new hypernyms at test time (i.e. *All organisms ate - All sheep ate* [ent.]). For convenience, we used four broad parent hypernyms, namely *organisms*, *microorganisms*, *produce*, *food* and *objects*. A complete overview of the lexical categories employed in the diagnostic experiment can be found in Table A.3 in Appendix A. Under these conditions, both GPT-2 and BERT still achieved a 99.9% accuracy. This finding suggests that the two models do not rely on memorization to solve the task. We assume that if they merely exploited a shortcut comprised of the operator, the hypernym and the corresponding label (e.g. *All animals* = entailment), they should fail in this test. Since this was not the case, we interpret this outcome as positive evidence that BERT and GPT-2 captured more holistic information about the 12 patterns as well as treated hypernymy as a

feature, as both were necessary prerequisites to succeed in this test.

Error Analysis

We conducted an error analysis on the few misclassified predictions made by the two models on the main evaluation of this split (284 errors for GPT-2 and 198 for BERT out of ∼468k test examples). A complete summary of these errors is provided in Table 5.10. The error analysis revealed that both models tended to misclassify premise-hypothesis samples that involve specific lexical pairs. As shown in Table 5.10, for GPT-2, many errors involved premise–hypothesis pairs containing hypernym-hyponyms such as *shoes–sneakers, instruments–organs, dinosaurus–allosaurus* and *stores–hardwarestores*. A similar pattern is observed for BERT, whose errors concentrated on lexical pairs involving the word *amphipods*. Although we cannot draw definite conclusions regarding the cause of these errors, we hypothesize that factors such as semantic ambiguity (e.g. *organ*) or frequency (e.g. *allosaurus, amphipods*) may have played a role. For instance, the word *organ* is lexically ambiguous, as it may refer either to a musical instrument or to a body part. Moreover, words such as *allosaurus* or *amphipods* are relatively infrequent, domain-specific terms that might be absent from GPT-2's vocabulary. As for the pair *stores–hardwarestore*, it should be noted that *hardware store* is conventionally written as a two-word expression (i.e. hardware store) rather than as a single compound.[3] It is thus possible that GPT-2 may be sensitive to the non-standard orthographic form used in the dataset. Interestingly, in both models' predictions, none of the aforementioned error-prone pairs failed in a systematic manner. In particular, the same pair was often classified correctly under some rules and incorrectly under others. For instance, as seen in Table 5.10, the pair *instruments–organs* was misclassified by GPT-2 in all of its 30 occurrences[4] in Rule 7, where examples have the form [P] *The organs were used in the concert* [H] *The instruments were used in the concert* = [Ent.]. In contrast, the same pair was misclassified only once under Rule 1 (e.g. [P] *All instruments were used in the concert* [H] *All organs were used in the concert* = [Ent.]), but was nevertheless classified correctly in all remaining rules in which it appeared.

Model	Rule	Total Errors	Misclassified Pairs	Errors / 30
GPT-2	R1	18	shoes–sneakers	17
			instruments–organs	1
	R4	21	shops–hardwarestores	8
			stores–hardwarestores	8
			arachnids–harvestmans	5
	R5	41	sneakers–footwear	24
			organs–instruments	17
	R6	15	vertebrates–alligators	6
			dinosaurs–allosaurus	4
			reptiles–alligators	3
			animals–alligators	2

[3]We verified the standard spelling of *hardware store* using the Cambridge (`https://dictionary.cambridge.org/dictionary/english/hardware-store` and Collins `https://www.collinsdictionary.com/dictionary/english/hardware-store`) dictionaries.

[4]Each lexical pair occurs 30 times within each rule (as it participates in 30 distinct sentence templates) and therefore 360 times across the entire test set.

	R7	188	instruments–organs shops–hardwarestores stores–hardwarestores vertebrates–alligators reptiles–alligators buildings–hardwarestores animals–alligators dinosaurs–allosaurus buildings–apartments footwear–sneakers	30 30 30 30 26 25 12 3 1 1
BERT-base	R2	71	arthropods–amphipods crustaceans–amphipods invertebrates–amphipods buildings–clothingshops	29 26 12 4
	R4	2	buildings–clothingshops stores–clothingshops	1 1
	R5	20	allosaurus–dinosaurs	20
	R6	64	crustaceans–amphipods arthropods–amphipods invertebrates–amphipods	28 26 10
	R8	41	arthropods–amphipods crustaceans–amphipods invertebrates–amphipods	21 15 5

Table 5.10: Summary of GPT-2's and BERT's errors on the Hyponym Generalization Split. "Total Errors" denotes the total number of misclassified premise-hypothesis pairs (out of 16,680) within each rule. "Misclassified Pairs" denotes the number of hypernym–hyponym pairs that were misclassified at least once out of the 556 pairs contained per rule. "Errors / 30" refers to the number of lexical pairs that were misclassified in all of their 30 occurrences within a rule. Rules with no errors are not included in this table.

5.2.2 Evaluation of the Hyponym-Hypernym Generalization Split

In our Hyponym–Hypernym Generalization Split, we eliminate the lexical overlap factor to obtain a clearer picture of what might drive our models' performance. As seen in Tables 5.6 and 5.11, the train and test data of this split contain completely disjoint hypernyms, hyponyms and sentence templates. In this respect, we have created this test in a way that the correct solution requires the models to capture and apply the patterns underlying the 12 monotonic rules, as there is no apparent lexical cue that can be exploited. We define a pattern as the combination of (i) the quantifier, (ii) the direction of the hypernym–hyponym substitution, and (iii) whether the substitution occurs in the premise or the hypothesis. Together, these factors determine the correct inference label. For example, the models must learn that in the case of Rule 1, under the quantifier *all*, a hypernym in the premise followed by a hyponym in the hypothesis yields an entailment relation (i.e. [P] all + hypernym [H] all + hyponym = [Ent.]). If, instead, models do rely strongly on the specific hypernyms or hyponyms observed during training (i.e. memorize *All + [P] animals + [H] horses = ent.* instead of the more generalized form *All + [P] hypernym + [H] hyponym* = [Ent.]) they should no longer be able to perform well in this test.

The results presented in Table 5.7 show that GPT-2 and BERT achieved an accuracy of 87.4% and 96.6% respectively under these conditions. Since the overlap in hyponyms and sentences is completely absent from this evaluation, we believe that this finding proves more clearly that GPT-2 and BERT pick up the monotonic patterns contained in this split, but also encode information about hypernym-hyponym relations. The comparatively better performance

Hyponym-Hypernym Generalization Split	
Train	**Test**
R1 All animals ate → All horses ate [E]	**R1** All mammals run → All rabbits run [E]
[P] all + hypernym [H] all + hyponym = Ent.	[P] all + hypernym [H] all + hyponym = Ent.
[...]	[...]
R4 Some animals ran → Some poodles ran [N]	**R4** Some toys broke → Some legos broke [N]
[P] some + hypernym [H] some + hyponym = Neutr.	[P] some + hypernym [H] some + hyponym = Neutr.
[...]	[...]
R12 Some roses bloomed → No flowers bloomed [C]	**R12** Some vehicles crashed → No cars crashed [C]
[P] some + hyponym [H] no + hypernym = Contr.	[P] some + hyponym [H] no + hypernym = Contr.

Table 5.11: Illustration of the Hyponym-Hypernym Generalization Split.

of BERT also suggests that this model type might have a better grasp of lexical relations or that it captured a more generalizable representation of monotonic patterns than GPT-2. We hypothesize that BERT's ability to encode input text in deep bidirectional representations might have enabled it to learn richer information about the premise-hypothesis pairs seen in training, contributing to this performance advantage. GPT-2, on the other hand, exhibited a performance drop of 12.5% compared to the previous two experiments. Although the two evaluation settings are not directly comparable, this small performance drop suggests certain interesting directions that could be investigated further in future experiments. First, the present split kept templates separate between the train and test data (e.g. Some animals run - Some poodles run vs. Some toys broke some legos broke). Since particularly GPT-2 generalized less effectively on disjoint templates, we assume that this is a dimension that would be worth-investigating in the previous split as well. Second, the depth of the lexical relations emerges as another possible contributing factor. In the previous diagnostic experiment, we replaced hypernyms with terms located higher in the hierarchy (e.g. animals, dogs, cats → *organisms*, fruit, vegetable → *produce*, toys, clothes → *things*), so the test pairs involve cases such as *beagle* ⊂ *organism*, *apple* ⊂ *produce*, *doll* ⊂ *thing*. In the present experiment, the hypernym-hyponym pairs remain closer in the hierarchy (e.g. *beagle* ⊂ *dog*, *apple* ⊂ *fruit*, and *doll* ⊂ *toy*). It is thus possible that the semantic distance between the parent-child concepts influences the difficulty of the task and, consequently, the models' performance, yet additional experiments that control for this factor would be needed to evaluate this hypothesis.

Error Analysis

In this setting, the number of errors increased for both models. We assume that two main factors may be responsible behind these misclassifications. First, it could be the case that individual monotonic rules might be more difficult for the two model types. Second, specific hypernym–hyponym pairs could be less robustly encoded than others, and therefore result in an increased number of errors. This is consistent with existing evidence, suggesting that hypernym–hyponym relations are generally well encoded, but not perfectly or uniformly (Ravichander and et al., 2020), while factors such as word frequency have been found to affect the strength of their encoding (Hanna and Mareček, 2021). Concerning the first factor, our zero- and few-shot experiments already showed that GPT-2 seems to be struggling more with rules that result in an entailing label, but predicts contradiction cases more accurately. As shown in Table 5.12, this pattern emerges in this test as well. Contradiction rules (i.e. Rules 9-12) scored an 100% accuracy by GPT-2, while entailment is the class that consistently contains the majority of misclassifications by GPT-2. BERT, on the other hand, exhibits a more uniform performance across labels and rules.

It should be noted that the test set of this split contains 556 unique hypernym–hyponym pairs, with each participating in 30 sentence templates. This means that each pair appears in 30 premise-hypothesis pairs per rule and 360 times across all 12 monotonic rules. In Tables 5.13

Model	R1	R3	R5	R7	R2	R4	R6	R8	R9	R10	R11	R12
	Entailment				Neutral				Contradiction			
GPT-2	75.3	58.2	76.0	76.7	89.0	94.2	89.5	90.0	100.0	100.0	100.0	100.0
BERT-base	97.2	88.7	99.3	99.0	97.8	99.6	94.1	95.3	89.4	100.0	100.0	99.4

Table 5.12: Per-rule accuracy of GPT-2 and BERT-base on the Hypernym–Hyponym Generalization test split.

Hypernym	Hyponyms	Rules
shops	toyshop, clothingshop, fruitshop, fishmonger, jewelryshop, antiqueshop, vegetableshop, drugstore, hardwarestore, petshop, bakery, bookshop, coffeeshop, cornerstore, supermarket, market	R1–8
arachnids	vinegaroon, spider, scorpion, harvestman, pseudoscorpion, opilione,tarantula, uropygi, amblypygi, ricinulei, solifugae,	R1–8
marsupials	antechinus, numbat, wombat, wallaby, bandicoot, bettong, bilby, dunnart, glider, opossum, phascolonus, quokka, quoll, ringtail	R1–8
mollusks	oyster, limpet, cockle, mussel, slug, abalone, snail, nautilus, octopus, scallop, cuttlefish, chiton, clam, squid	R1–8
shrubs	potentilla, buckthorn, dogwood, elder, heather, snowberry, weigela, viburnum, rose, laurel, winterberry, cotoneaster, clethra, currant, barberry, cassia, azalea, gooseberry, gorse, lavender	R1–8
headwear	helmet, beanie, hairband, headband, headscarf, balaclava	R1–8
cetaceans	dolphin, whale, sei,orca, beluga, narwhal	R1–8
viruses	pestivirus	R1–8
dogs	mudi, greyhound	R1–8

Table 5.13: Hypernym–hyponym pairs that were misclassified completely (30/30 times per rule) across multiple rules by GPT-2 in the Hyponym–Hypernym Generalization Split. Hyponyms highlighted in brown denote lexical pairs that also appear in BERT's misclassifications.

and 5.14 we have included those hypernym–hyponym pairs that were misclassified in all 30 of their occurrences within multiple monotonicity rules by GPT-2 and BERT. As illustrated in Table 5.15, again no pair failed systematically in neither models' predictions. In particular, we observe that a lexical pair could be misclassified in all 30 sentence templates within a particular monotonic rule or only in a subset of the templates. For example, under Rule 1, GPT-2 produced errors for 154 of the 556 lexical pairs, but only 120 of these pairs were misclassified in all 30 premise-hypotheses where they appeared. This indicates that the remaining 34 pairs were classified correctly in at least some sentence templates. At the same time, a pair that consistently failed under one rule could still be classified correctly under another, indicating that no lexical pair failed completely across the entire dataset. As shown in Table 5.13, GPT-2 mislabeled a considerable number of inferences containing specific hypernym-hyponym pairs, across non-contradiction rules. Similarly to the previous experiment, many of these error-prone pairs involve rare, domain-specific terms (e.g. *uropygi, viburnum, and potentilla*), as well as non-standard compound forms, such as *vegetableshop* instead of *vegetable shop* and *clothingshop* instead of *clothing shop*. On the other hand, GPT-2 also made errors on highly frequent lexical pairs, such as *shops* $\supset$ *supermarket*. Interestingly, these error-prone pairs were classified correctly when they appeared in inferences resulting in a contradiction label. Although the cause of this behaviour remains unclear, we hypothesize that if contradiction rules are indeed solved through surface-level cues associated with the presence of negation words, then the difficulty of a lexical relation may play a less important role, as the shortcut alone may be sufficient to guide the model to the correct prediction. As illustrated in Table 5.14, BERT exhibits a smaller number

Hypernym	Hyponyms	Num. of Rules
arachnids	spider	9
arachnids	pseudoscorpion, schizomida, opilione, scorpion, tarantula, vinegaroon, harvestman	4
cetaceans	whale	9
cetaceans	sei, dolphin	4
shops	toyshop, clothingshop,	5

Table 5.14: Hypernym–hyponym pairs that were misclassified frequently (30/30 times per rule) in many rules by BERT-base on the Hyponym–Hypernym Generalization Split. The Number of Rules column indicates the number of rules where a given pair was misclassified completely.

Rule	Total Errors (out of 16,680)		Pairs with Errors (out of 556)		Pairs with 30/30 Errors	
	GPT-2	BERT	GPT-2	BERT	GPT-2	BERT
R1 (All animals ate → All horses ate)	4119	459	154	64	120	4
R2 (All horses ate → All animals ate)	1832	358	76	29	48	6
R3 (Some poodles ran → Some dogs ran)	6957	1882	256	110	189	22
R4 (Some dogs ran → Some poodles ran)	961	66	53	5	24	1
R5 (No shops closed → No bakeries closed)	3988	115	154	9	117	2
R6 (No bakeries closed → No shops closed)	1745	969	72	61	39	20
R7 (The toys broke → The legos broke)	3883	163	153	16	104	2
R8 (The legos broke→ The toys broke)	1661	775	67	61	35	13
R9 (All birds flew → No sparrows flew)	0	1757	0	130	0	9
R10 (No sparrows flew → All birds flew)	0	0	0	0	0	0
R11 (No vehicles crushed → Some cars crashed)	0	0	0	0	0	0
R12 (Some cars crashed → No vehicles crashed)	0	93	0	6	0	1

Table 5.15: Summary of GPT-2's and BERT-base's errors across the monotonicity rules in the Hyponym–Hypernym Generalization Split. "Total errors" denotes the total number of misclassified premise-hypothesis pairs (out of 16,680) within each rule. "Pairs with errors" denotes the number of hypernym–hyponym pairs that were misclassified at least once out of the 556 pairs contained per rule. "Pairs with 30/30 errors" refers to the number of lexical pairs that were misclassified in all of their 30 occurrences within a rule.

of systematically misclassified pairs than GPT-2, which are mostly concentrated around the hypernyms *arachnids, cataceans* and *shops*. We further notice that there is a high degree of overlap between the errors of the two models, suggesting that some semantic relations are challenging for both model types.

5.2.3 Evaluation of the Flipped Pattern Split

Our previous two experiments showed that when BERT and GPT-2 are trained on all monotonicity patterns contained in MoRe, and therefore observe how hypernym–hyponym relations interact with monotonicity, they can successfully solve inferences of the same quantifier structure involving unseen lexical relations. However, so far, we have evaluated our models on inferences that always preserve the same underlying structure as the training data. Even when we kept hypernyms and hyponyms disjoint, the pattern-label mapping between the train and the test data always remained constant. For instance, a train instance such as *All animals ate → All horses ate* and a test instance such as *All mammals run → All rabbits run* instantiate the same underlying pattern: *[P] all + hypernym [H] all + hyponym = Entailment*. In this sense, we manipulated the train and test data in a way that allowed the two models to succeed by applying the same patterns that had be learnt during fine-tuning.

In the Flipped Pattern Split, we create an extreme diagnostic setting, to demonstrate what

Train Shortcut hypernym$_P$ → hyponym$_H$	**Test Shortcut** hyponym$_P$ → hypernym$_H$
R1 All animals ate → All horses ate [E]	**R2** All horses ate → All animals ate [N]
R4 Some animals ran → Some poodles ran [N]	**R3** Some poodles ran → Some animals ran [E]
R5 No shops opened → No bakeries opened [E]	**R6** No bakeries opened → No shops opened [N]
R7 The shoes were wet → The sneakers were wet [E]	**R8** The sneakers were wet → The shoes were wet [N]

Table 5.16: Illustration of the Flipped Pattern split.

happens when this form of pattern matching is no longer sufficient to solve the task. Specifically, assuming that both models capture training patterns closely, in this split we introduce an intentionally misleading shortcut in the training data. As seen in Table 5.16, in training we only include rules that that teach a model the association *"operator + **hypernym** in the premise + **hyponym** in the hypothesis = entailment / neutral"*. At test time, however, we eliminate the training shortcut, by evaluating models on rules that instantiate the exact opposite pattern: *"operator + **hyponym** in the premise + **hypernym** in the hypothesis = entailment / neutral"*. As discussed in the previous chapters, since the training data reinforces this strong statistical association, and fine-tuned models generally tend to exploit the easiest solution available in a training distribution (Geirhos et al., 2020), we have no reason to expect the two models to reverse this dominant learning signal. The results presented in Table 5.7 show that, in line with our expectation, both BERT-base and GPT-2 achieved a 0% accuracy under this condition, which suggests that the two models indeed captured and consistently applied the shortcut that was deliberately embedded in training. In practice, this means that for every flipped rule in the test set (i.e. R2,R3,R6,R8), both GPT-2 and BERT-base predicted the label corresponding to their training rule counterpart (i.e. R1,R4,R5,R7) across all instances. For instance, R2 (e.g. All horses ate → All animals ate) was consistently predicted as entailment, given that its training equivalent, R1 (e.g. All animals ate → All horses ate), results in this label, even though this inference relation no longer holds in R1's flipped version.

It should be noted that our Flipped Pattern experiment constitutes a *concept shift* generalization test. Concept shift is a demanding type of distributional shift where the feature-label relationship changes at test time, and is generally viewed as a generalization setting where LLMs struggle considerably (Budnikov et al., 2024; Nguyen et al., 2025; Lyu et al., 2026). In our case, the individual constituents of the monotonic patterns are shared between the train and test split. In particular, the models are exposed to the same quantifiers, they have seen both the entailment and neutral label as well as multiple hypernym–hyponym pairs during training. Yet, to succeed in this test, a system should in theory recombine these familiar features to infer a previously unseen input–label mapping. This, however, constitutes a form of compositional, systematic reasoning that, as discussed in Chapter 1 (Section 2.3.1), neural networks struggle to exhibit (Pavlick, 2022; Goodwin et al., 2020; Bender et al., 2021; Vegner et al., 2025), and finetuning has not convincingly induced (Zhang et al., 2022; Pavlick, 2022; Chen and Gao, 2023). On the other hand, because the shortcut is perfectly predictive during training and completely invalid at test time, the models are essentially given no opportunity to overcome it. For the purposes of this thesis, we decided to adopt this extreme scenario to isolate the effects of shortcut learning as clearly as possible. In line with prior work on shortcut learning (Bastings et al., 2022; Eshuijs et al., 2025), a natural extension of this setup would be to investigate whether a model can overcome a biased signal. One idea would be to include shortcuts only in a proportion of the training data of this split, yet, we leave this as an interesting direction for future work.

Chapter 6

Conclusion and Discussion

6.1 Conclusions

The objective of this thesis was to investigate how two models, GPT-2 and BERT-base handle the phenomenon of Monotonic Reasoning within the broader task of Natural Language Inference. Given that foundation models such as GPT-2 and BERT-base are not explicitly designed to perform formal reasoning processes , and that the predictions of fine-tuned NLI models have often been found to deviate significantly from the theoretical goal of principled linguistic reasoning, we treated monotonicity as a diagnostic setting to investigate which specific components of this phenomenon can be acquired through supervised learning, what linguistic information must be available during training, and where the limits of such learning lie.

The most straightforward way to investigate what information GPT-2 already possesses about the task was through our in-context-learning experiments. In line with our expectations, we take the near-chance performance that both zero- and few-shot GPT-2 reported on SNLI and MoRe as evidence that the concept of NLI is too demanding to emerge spontaneously from GPT-2's pre-training objective. A key observation is that fine-tuning compensated for GPT-2's limited pre-trained knowledge of monotonic reasoning, as reflected in the sharp performance gap between its zero-shot (33.3%) and fine-tuning (99.9%) accuracy on the main test split of MoRe. This finding proves the effectiveness of supervised fine-tuning in aligning a model's behavior with the demands of a given task. Moreover, despite much recent evidence surrounding the potential of LLMs as few-shot learners (Brown et al., 2020), our findings showed that adding labeled premise-hypothesis demonstrations in the prompt proved insufficient to specialize the model for monotonicity. Previous studies confirm that even larger few-shot models such as FLAN-T5 and GPT-3 perform near-random chance on monotonic inference tasks in standard prompting settings (Chen and Gao, 2023). For this reason, we hypothesize that more advanced techniques are likely necessary to induce more robust inference behaviors, some of which we discuss in Section 6.2.

The second question we addressed is the extent to which training on a more general and diverse NLI benchmark can help GPT-2 generalize to monotonic inferences. Consistent with prior findings, our SNLI fine-tuned GPT-2 reported a random-chance accuracy of 34,6% on MoRe, reflecting the well-documented tendency of NLI models to generalize poorly across different types of inference (Geiger et al., 2020; Rozanova et al., 2021, 2022; Chen and Gao, 2023; Wijnholds, 2023). One possible hypothesis is that SNLI does not contain a sufficient amount of monotonic samples to enable a better transfer between the two datasets. As noted by Geiger et al. (2020), SNLI includes premise-hypothesis pairs containing hypernyms and hyponyms, which means that the dataset provides at least a partial learning signal about a core linguistic component of monotonic logic. However, this information alone did not seem to help GPT-2 handle the more complex quantifier logic of monotonicity. Another way of interpreting this finding relates to the broader problems that LLM's face when they learn to "reason" from data. It is generally accepted that fine-tuning does not induce a form of abstract reasoning-like abilities that could in principle allow a system to generalize flexibly across datasets that contain varia-

tions of the same phenomenon (Chen and Gao, 2023). If that was the case, we might expect a better out-of-distribution performance, considering that both SNLI and MoRe instantiate the same task and share the same balanced label space (i.e. entailment,neutral, contradiction). We hypothesize that augmenting SNLI with instances from MoRe could improve generalization on this test, yet due to time constraints we did not investigate this possibility.

The most central part of our methodology involved the experimental splits of MoRe. As explained in previous chapters, there are many open questions in the literature regarding how fine-tuning affects the linguistic knowledge captured in a model's representations and the extent to which appropriate features are actually leveraged at test time (Geirhos et al., 2020; Mosbach et al., 2020; Talmor et al., 2020; Zhang et al., 2022). In this work, we approached this question on a behavioral level, and therefore designed these carefully manipulated train/test splits to investigate if two specific linguistic features, namely hypernymy and logical operators, are captured and employed by our fine-tuned models. On a secondary level, some our tests aimed to isolate and expose potential shortcut strategies underlying their predictions.

The evaluation of our experimental splits revealed positive behavioral evidence about the use of hypernymy as a feature. As discussed in Sections 5.2.1 and 5.2.2, both GPT-2 and BERT-base achieved a near-perfect accuracy when they were fine-tuned and tested on familiar monotonic patterns that contained unseen hyponyms (99.9% for both models) and hyponym-hypernym pairs (87.4% and 99.6% respectively). This high performance on both splits seems to indicate that the models encoded and learnt to use hypernymy information quite well, considering that this was a necessary prerequisite to solve the two tests. In fact, both models performed more robustly than expected. We showed that even when a very strong lexical-overlap heuristic was introduced in the Hyponym Generalization split, neither model reported a drop after the shortcut was removed. This implies that even though GPT-2 and BERT could exploit this shallow, yet perfectly predictive cue, they likely captured something more nuanced about the structure of our examples. Nevertheless, the fact that our fine-tuned models overcame this particular heuristic does not rule out the possibility that other spurious strategies may have taken place. Considering that the search space for shortcuts is vast (or even unbounded), and a dataset cannot be fully free of biases (Linzen, 2020), it is difficult to control for every undesired association a model might exploit. Prior work has identified various unsystematic behaviors behind NLI models' decision making, ranging from treating individual words as proxies for a label (Gururangan et al., 2018), to making predictions without considering the entire premise-hypothesis pair (Poliak et al., 2018; Glockner et al., 2018). Given this background, we think that a more thorough exploration of possible heuristics in these tests would be worth investigating in future work.

More broadly, our analysis showed that GPT-2 and BERT-base generalized successfully only in those tests that kept monotonic patterns stable at test time. This observation seems to reflect the view that fine-tuned models rely on pattern-matching behaviors to simulate inference (Bender and Koller, 2020; Jiang et al., 2024; Shojaee et al., 2025). This means that they exploit *surface-level* associations between the inputs and labels of a given distribution, and use them to predict outputs. The underlying hypothesis about a system that operates via pattern-matching is that it will generalize successfully as long as the learnt associations extend, but break when these associations are disrupted (Geirhos et al., 2020; Makar et al., 2022). Our Flipped Pattern split was designed as an extreme, yet useful diagnostic setting to test precisely this hypothesis. We showed that when a general system is trained on a fully biased shortcut signal, it has no way to decipher if this particular association is spurious or meaningful to solve the test. In this sense, the 0% accuracy reported in this test should not be interpreted as our models' failure to learn. Fine-tuned LLMs are predisposed to learning patterns. Since the surface features of this split (i.e. *operator + hypernym , operator + hyponym*) correlated always with a given label (i.e. *entailment / neutral*), our models performed as expected: they captured and employed these associations, and this is precisely the type of behavior that fine-tuning was expected to elicit.

One view in the literature is that shortcut-learning is an inherent frontier of the probabilistic LLM paradigm (Peters et al., 2015; Geirhos et al., 2020; Shah et al., 2020; Makar et al., 2022; Zhang et al., 2022; Hermann et al., 2024; Sun et al., 2024). Since general systems lack robust mechanisms to abstract logical principles from examples (e.g. $All\ x \Rightarrow All\ y$ when $x \subset y$)

and manipulate their components in new situations (e.g. *All* $y \not\Rightarrow$ *All* x), much recent work deviates from standard, data-centric fine-tuning, and explores alternative methods to enhance LLM inference with a more explicit modeling of reasoning processes. In the next section, we discuss some of these approaches, but also highlight the limitations of the present work, and suggest possible directions for future research.

6.2 Limitations and Future Research

The methodology employed in this thesis could improve in several ways, but at the same time provides many opportunities for future work. From a more technical perspective, due to computational and time constraints, we did not conduct an exhaustive hyperparameter search in our fine-tuning experiments. We therefore consider this a factor that could be further explored. A second aspect concerns our choice of models. BERT-base and GPT-2 were selected as they are relatively small, well-studied models that met our compute requirements and allowed us to situate our findings within the existing NLI literature. Yet, it would be interesting to investigate if larger, or more recent reasoning models like DeepSeek-R1 (Guo et al., 2025) or OpenAI's o1 (OpenAI, 2024) exhibit greater potential for monotonicity reasoning than smaller, conventional LLMs, particularly under in-context learning conditions.

For the scope of this thesis, we employed standard training techniques such as supervised fine-tuning and in-context-learning. Since our objectives were diagnostic, we believe that these methods were a sufficient first step towards this end. Future work using our dataset could investigate if more sophisticated methods can result in more robust performance on monotonic reasoning. One popular technique is Chain-of-Thought (COT) prompting. There is evidence that asking LLMs to explain their intermediate reasoning steps before predicting a label leads to better performance on a range of logical reasoning tasks (Kojima et al., 2023), but also makes them less dependent on shortcut learning in NLI Yuan et al. (2024). CoT has also been recently integrated in Reinforcement Learning frameworks, surpassing finetuned models on general NLI datasets such as SNLI and MNLI (Miralles-González et al., 2025). There is also an increasing interest for other prompt-based approaches such as prefix-tuning (Li and Liang, 2021), and P-Tuning (Liu et al., 2023b) which have also been found to be more effective than fine-tuning on the task of NLI. Prompted by the fundamental limitations of neural NLI, recent work has also attempted to augment LLMs with explicit symbolic representations, giving rise to hybrid neuro-symbolic systems (Liu et al., 2022; Creswell et al., 2022; Farjami et al., 2026) that have already shown promising potential in Monotonic NLI Chen and Gao (2023).

Additionally, we acknowledge that since our experiments were only behavioral, our evidence could be strengthened through complementary mechanistic or representational analyses. For instance, the high-accuracy reported on our Hyponym and Hyponym-Hypernym Generalization splits points strongly towards the interpretation that the models picked up the given monotonic patterns and learnt to encode hypernymy information. However, as highlighted in previous work, a behavioral evaluation cannot by itself demonstrate that a model possesses an internal representation of a particular linguistic feature (Geiger et al., 2020). Future work could therefore employ probing classifiers to reveal how well models encode information about quantifiers or hypernymy internally. In line with prior Monotonic NLI studies (Geiger et al., 2020), causal intervention methods could further show if models compute something more nuanced about how these two features interact. These supplementary experiments could also help us clarify why some rules were more error-prone than others, as well as why certain lexical pairs appeared frequently among the errors of both models.

Overall, we designed MoRe as a highly controlled and targeted diagnostic resource that allows for a wide range of manipulations. For this reason, we view our experimental splits as only the first step in exploring our dataset's potential, and thus encourage future studies to extend and explore it further. For instance, some interesting directions would be to control for additional linguistic factors (e.g. sentence length, word frequency, add distractors, insert new semantic relationships like synonymy or antonymy), as well as to integrate it more diverse training and evaluation frameworks (e.g. mechanistic interpretability, data augmentation, ad-

versarial testing). As noted so far, we believe that one particularly promising application of MoRe is the study of shortcut learning. This motivated the design of the Flipped Pattern split, which allowed us to inspect the effect of an extreme shortcut on our model's performance. An interesting extension of this setup would be to investigate whether and under which conditions a model may overcome this shortcut association. In line with prior work (Bastings et al., 2022; Eshuijs et al., 2025; Suhail et al., 2025), one option would be to introduce a misleading signal only in a proportion of the training data. This could illustrate if models prioritize useful vs. spurious features when the latter are no longer consistently predictive of class labels. Beyond diagnosing shortcut reliance behaviorally, recent work on Mechanistic Interpretability investigates whether shortcut mechanisms can be detected in model internals through methods such as logit attribution and path patching (Eshuijs et al., 2025), as well as proposes ways of mitigating models' reliance on such heuristics (Du et al., 2021; Eshuijs et al., 2025). We believe that employing similar methods in the context of the Flipped Pattern split would provide valuable complementary evidence to our behavioral findings, and hope to see future research build on this idea.

Appendix A

Additional Tables



Figure A.1: Confusion matrix for GPT-2 fine-tuned on SNLI and evaluated on the Hyponym Generalization Split of MoRe.

Premise	**Hypothesis**
He knows neither us.	He knows neither of us.
I know neither of his friends.	I know neither of his brothers.
He knows neither of us.	He knows neither of letters of the alphabet.
I know neither of the two members.	I know neither of the two sisters.
There is no open fireplace without smoke.	There is no fire without smoke.
There is no hope of his coming back safe.	There is no hope of his coming back birth control device.
It is no use blaming him for the accident now.	It is no use blaming him for the accident nowaday.
There is no life without electricity and water.	There is no period without electricity and water.
Since it's not the rainy season, there's no need for an umbrella.	Since it's not the rainy season, there's no need for an union.
I gave him a warning, but he took no notice of it.	I gave him a warning, but he took no notice of engineering science.
These books are accessible to all students.	These books are accessible to all nonachievers.
I thank you with all my heart.	I thank you with all my two-dimensional figure.
There is no such thing as an essential.	There is no such thing as a must.
There is no link between these two.	There is no link between these snake eye.
We have no need for relief.	We have no need for assistance.
I cannot water the backyard. There is no water.	I cannot seven sea the backyard. There is no seven sea.
Danny has no sense of example.	Danny has no sense of beauty.

Table A.1: Ill-formed examples collected from the HELP dataset.



Figure A.2: Confusion matrix for GPT-2 fine-tuned on SNLI and evaluated on the Hyponym Generalization Split of MoRe.

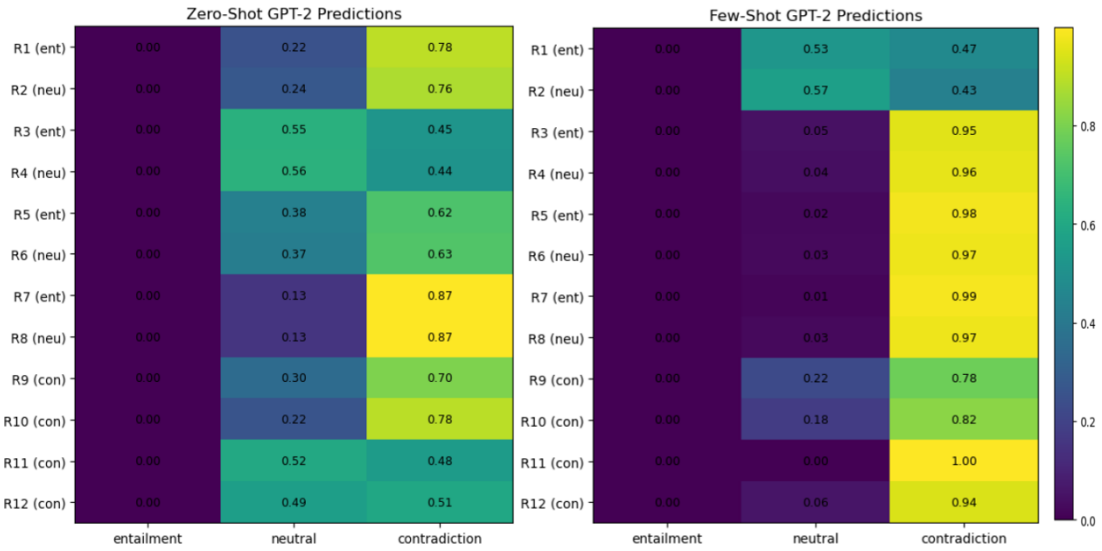


Figure A.3: Confusion Matrix for GPT-2's predictions under zero-shot and few-shot prompting.

	Hyponym Gen.				**Hypo-Hypernym Gen.**				**Flipped Pattern**			
Model	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1
GPT-2	99.9	99.9	99.9	99.9	87.4	88.2	87.4	87.3	0.0	0.0	0.0	0.0
BERT-base	99.9	99.9	99.9	99.9	96.6	96.7	96.6	96.7	0.0	0.0	0.0	0.0

Table A.2: Classification Report for GPT-2 and BERT-base on MoRe's Experimental Splits

New hypernyms	**Original hypernyms**
organisms	animals, vertebrates, invertebrates, arthropods, mammals, dogs, canines, primates, rodents, felines, ungulates, marsupials, cetaceans, birds, fish, reptiles, amphibians, insects, arachnids, crustaceans, mollusks, dinosaurs, protozoa
microorganisms	bacteria, microbes, microorganisms, viruses, pathogens
produce	fruit, vegetables
food	sweets, desserts
objects	accessories, appliances, clothing, devices, electronics, footwear, furniture, garments, headwear, instruments, jewellery, shoes, tops, toys

Table A.3: This table presents the superordinate terms introduced in the diagnostic evaluation of the Hyponym Generalization Split, and the original hypernyms mapped to each of them.

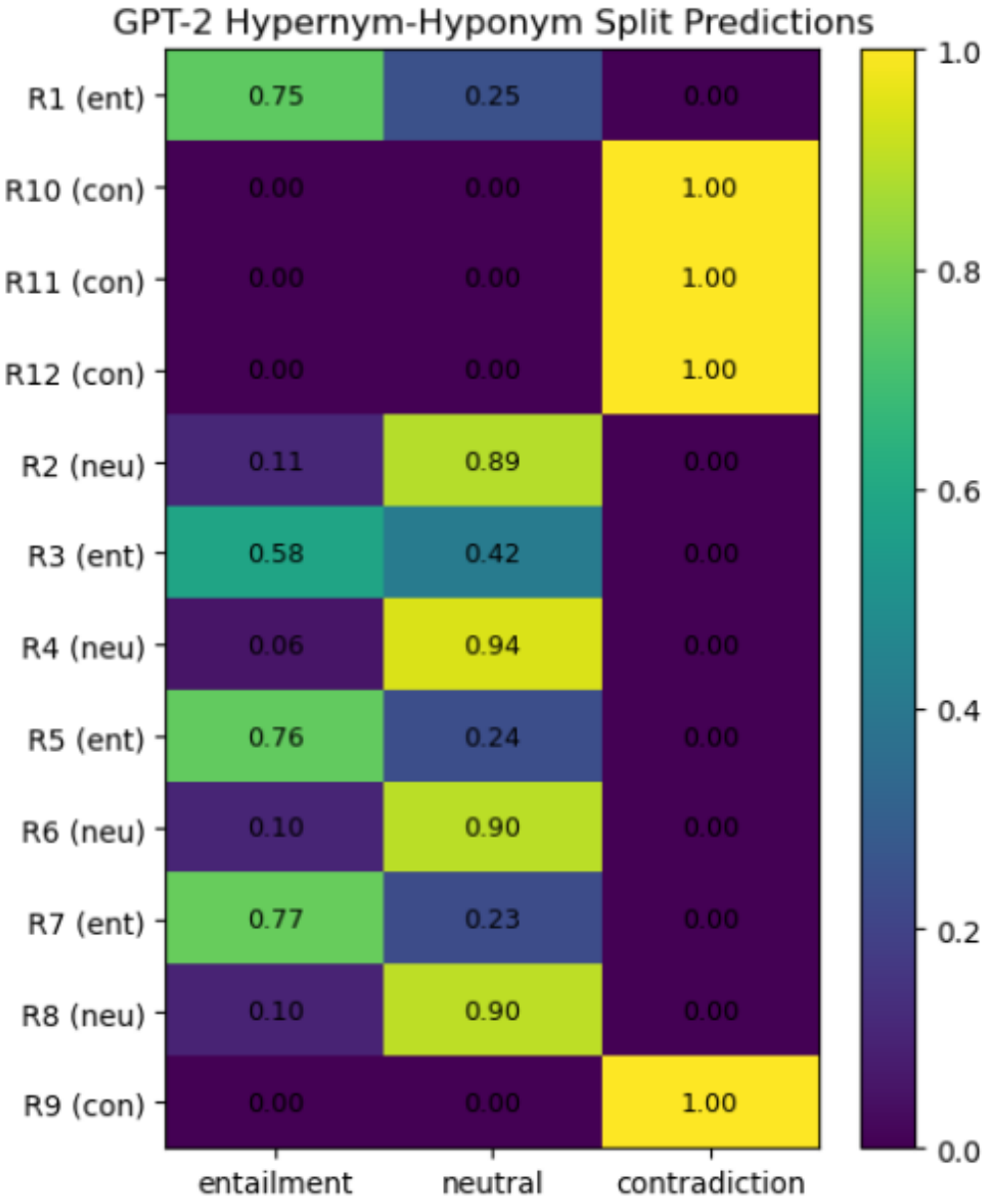


Figure A.4: Confusion Matrix of GPT-2's predictions on the Hypernym-Hyponym Split of MoRe

Train Shortcut	Test Rules	Acc.
R1: All animals ate → All horses ate [E] hypernym [H] + hyponym [P] = Entailment	**R3**: Some horses ate → Some animals ate [E]	0.0
R2: All horses ate → All animals ate [N] hyponym [H] + hypernym [P] = Neutral	**R4**: Some animals ate → Some horses ate [N]	
R1: All dogs run → All beagles run [E] hypernym [H] + hyponym [P] = Entailment	**R5**: No dogs run → No beagles run [E]	100.0
R2: All beagles run → All dogs run [N] hyponym [H] + hypernym [P] = Neutral	**R6**: No beagles run → No dogs run [N]	
R1: All birds flew → All sparrows flew [E] hyponym [H] + hypernym [P] = Entailment	**R3**: Some sparrows flew → Some birds flew [E]	100.0

Table A.4: Overview of the additional shortcut-introducing experiments conducted with combinations of MoRe's Rules. The accuracy column shows GPT-2's and BERT's performance on these Experiments.

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